



ELEMENTS OF MECHANICAL ENGINEERING LABORATORY MANUAL

I/II Semester (1BEMEL207)



Name of the Student :

Semester /Section :

USN :

Batch :

DAYANANDA SAGAR COLLEGE OF ENGINEERING

**Accredited by National Assessment & Accreditation Council (NAAC) with 'A' Grade
(An Autonomous Institution affiliated to Visvesvaraya Technological University, Belagavi
&**

ISO 9001:2008 Certified

**DEPARTMENT OF MECHANICAL ENGINEERING
SHAVIGE MALLESWARA HILLS, KUMARASWAMY LAYOUT
BENGALURU-560111**

(Backside of cover page)

Vision of the Institute

To impart quality technical education with a focus on Research and Innovation emphasizing on Development of Sustainable and Inclusive Technology for the benefit of society.

Mission of the Institute

- To provide an environment that enhances creativity and Innovation in pursuit of Excellence.
- To nurture teamwork in order to transform individuals as responsible leaders and entrepreneurs.
- To train the students to the changing technical scenario and make them to understand the importance of Sustainable and Inclusive technologies.



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VISION OF THE DEPARTMENT

To Prepare Mechanical Engineers having technical competency and Managerial Skills driven by human values and ignite the young minds capable of addressing ever-changing global issues by Research and Innovation.

MISSION OF THE DEPARTMENT

- **To** provide a platform that imparts Scientific Knowledge and Technical skills.
- **To** train students to demonstrate their Communication and Managerial skills with Ethical practices.
- **To** engage students in professional activities through Research, Higher Education and Life Long Learning.

PROGRAMME EDUCATIONAL OBJECTIVES [PEOs]

PEO-1: Graduates shall exhibit the knowledge and competency for successful career in Mechanical Engineering and pursue higher studies.

PEO-2: Graduates shall exhibit the necessary skills and manage to interact in the field of profession with different Cross Functional Groups.

PEO-3: Graduates shall demonstrate their Engineering Profession by adopting Ethical and Professional practices.

PEO-4: Graduates shall engage in Professional society interaction, training and research to enable Life Long Learning.

PROGRAMME SPECIFIC OUTCOMES [PSOs]

- **PSO1-** Graduate will be able to apply basic science, mathematics and engineering knowledge to conceptualize, model, design and manufacture physical systems and processes.
- **PSO2 -** Graduate will be able to utilize the knowledge of advanced engineering tools in design & thermal engineering domain.
- **PSO3-** Graduate will be able to lead professionally in an industrial environment by applying the management principles related to organizational planning, leadership, and research, production and service activities.

Elements of Mechanical Engineering Lab		Semester	II
Course Code	1BEMEL207	CIE Marks	50
Teaching Hours/Week (L: T:P:S)	0:0:2	SEE Marks	50
Total Hours of Pedagogy	24 Hours (12-week session)	Total Marks	100
Credits	1	Exam Hours	3
Examination type (SEE)	Practical		
Course Outcomes			
At the end of the course, the student will be able to:			
CO1: Perform basic workshop and machine shop operations such as arc welding, soldering, sheet metal fabrication, and simple lathe operations by following standard operating procedures and safety practices.			
CO2: Conduct experimental measurements and evaluations by calibrating common measuring instruments using standard gauges and determining physical properties of fuels and oils such as flash point, fire point, and viscosity.			
CO3: Apply fundamental principles of engineering mechanics to analyze force systems, moments, equilibrium, centroids, and second moment of area for simple mechanical systems and components through laboratory demonstrations and exercises.			
Note:			
1. The laboratory syllabus consists of PART-A and PART-B. While PART-A has 5 conventional experiments, PART-B has 5 typical open-ended experiments. The maximum marks for the laboratory course is100.			
2. Both PART-A and PART-B are considered for CIE and SEE.			
3. For continuous internal evaluation, during the semester, classwork, record book and CIE Test marks should be considered and to be evaluated for 50 marks			
PART – A			
CONVENTIONAL EXPERIMENTS			
1. Introduction to Mechanical Engineering Lab – Welding, Soldering, Sheetmetal work and Tools used.			
2. Preparation of welded joints using the arc welding process.			
3. Fabrication of a sheet metal part.			
4. Basic machining operations using lathe			
5. Calibration of measuring instruments using standard gauges			

PART – B

TYPICAL OPEN-ENDED EXPERIMENTS

Open-ended experiments are a type of laboratory activity where the outcome is not predetermined and students are given the freedom to explore, design, and conduct the experiment based on the problem statements as per the concepts defined by the course coordinator. It encourages creativity, critical thinking, and inquiry-based learning.

1. Comparative study of flash point and fire point of various fuels
2. Comparative study on viscosity of different base fuels.
3. Introduction to Engineering Mechanics, Forces, Vectors and Resultants, Moments and Couples, Two-dimensional equilibrium
4. Centroids and Centre of Gravity, Moment of Inertia- Second moment of Area
5. Rectilinear and Angular motion - Motion Analysis using Pick-and-Place robot

Suggested Learning Resources: (Textbook / Reference Book/ Manuals):

Textbooks:

1. Amitabh Ghosh and Amit Kumar Mallik, Manufacturing Science, Affiliated East West Press (p) Ltd, New Delhi, 2002
2. Hajara and Choudhary, Workshop Technology Vol. I (2008) & II (2010), Median Promoters & publishers, Bombay.
3. Khanna O. P, Workshop Practice, Vol. I, Dhanpat Rai & Co., 2000.
4. Applied Mechanics using Solidworks – Godfrey Onwubolu Imperial College Press, 2015
5. Engineering Mechanics - Statics and Dynamics by Vijaya Kumar Reddy K, Suresh Kumar J.BS Publications, 2024

Reference books / Manuals:

1. Serope Kalpakjian and Steven R Schmid, Manufacturing Engineering and Technology, Fourth Edition, Pearson Education, Asia, 2000.
2. P.N. Rao, Manufacturing technology--Foundry, Forming and Welding, Tata McGraw Hill Education, 2001.
3. Engineering Mechanics, S. Timoshenko, D. H. Young, J.V. Rao, McGraw. Hill Education 2017. 5th Edition.

Web links and Video Lectures (e-Resources):

- <https://www.millerwelds.com/resources/article-library/the-fundamentals-of-weldingprocess-equipment-and-applications>
- <https://www.youtube.com/watch?v=TIhGTSDfQxc>
- <https://www.coursera.org/learn/engineering-mechanics-statics>
- https://onlinecourses.nptel.ac.in/noc21_me70/preview

Teaching-Learning Process (Innovative Delivery Methods):

The following are sample strategies that educators may adopt to enhance the effectiveness of the teaching-learning process and facilitate the achievement of course outcomes.

- Flipped Classroom
- Simulation and Virtual Labs
- Video demonstration

Assessment Structure:

The assessment for each course is equally divided between Continuous Internal Evaluation (CIE) and the Semester End Examination (SEE), with each component carrying **50% weightage** (i.e., 50 marks each).

The CIE marks awarded shall be based on the continuous evaluation of the laboratory report using a defined set of rubrics. Each experiment report can be evaluated for 30 marks. The laboratory test (duration 03 hours) at the end of the last week of the semester /after completion of all the experiments (whichever is early) shall be conducted for 50 marks and scaled down to 20 marks. For both CIE and SEE, the student is required to conduct one experiment each from both Part A and Part B.

CO-PO Mapping

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PS01	PS02	PS03
CO1	2							2			1	1		
CO2	2							2			1	1		
CO3	3	2		1	3			2			1	1	2	

Scheme for Examination:

One Question from Part A	- 20 Marks	(05 Writeup+15)
One Question from Part B	- 20 Marks	(05 Writeup+15)
Viva-Voce	- 10 Marks	

Total	- 50 Marks	

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1	Introduction to Welding process	1
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9	Calibration of measuring instruments (Micrometer, Vernier caliper and Vernier height gauge)	36
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12	Introduction to Engineering Mechanics, Forces, Vectors and Resultants	51
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DO's

- Students must always wear uniform and shoes before entering the lab.
- Proper code of conduct and ethics must be followed in the lab.
- Windows and doors to be kept open for proper ventilation and air circulation.
- Note down the specifications/drawings before working on the preparation of models.
- Receive the tools and materials required for preparation of models with signing in register.
- Use of safety goggles/ face shield during welding.
- Do the models under the supervision/guidance of a lecturer/lab instructor only.
- Keep the sufficient distance from other students while preparing models.

DONT's

- Do not touch electrical circuits of welding machine.
- Don't gather while preparing models, that may hurt other with tools.
- Don't unlock snip/sheet metal cutter lock, without use.

WELDING

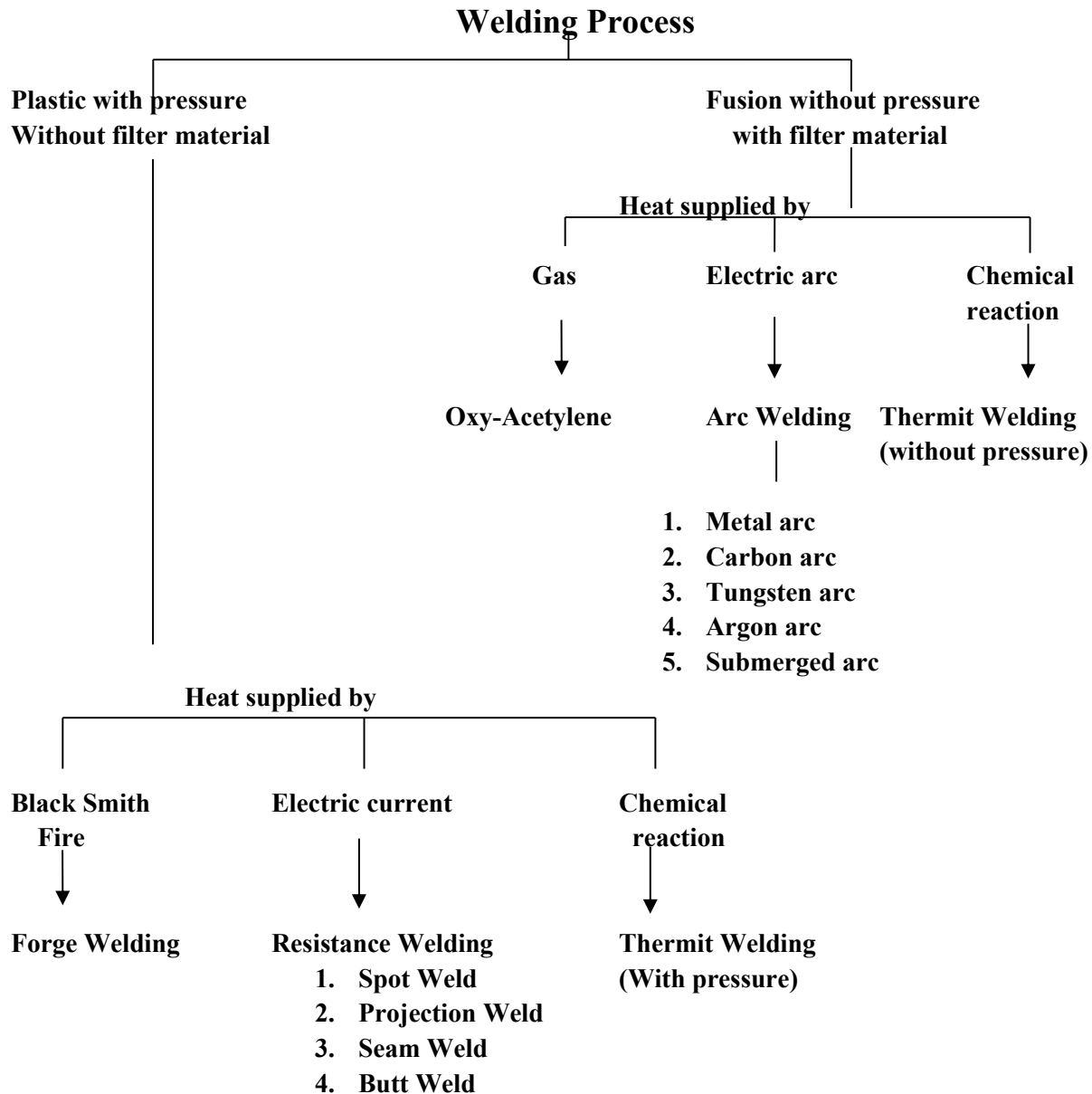
Welding is the metallurgical process of joining two metals by application of heat with or without pressure and filler material.

The **weldability** is defined as the capacity of being welded into inseparable joints having specified properties. If the chemical, physical, thermal and metallurgical properties are undesirable in welded joints then they can be corrected by using proper shielding atmosphere, fluxing material, proper filler material, proper welding procedure and heat treatment before and after welding. The following metals have good weldability in descending order. Iron, carbon steels, cast steels, cast iron, low alloy steels, and stainless steel.

Welding processes can be broadly classified under two broad sub classes.

- Plastic welding / pressure welding

- Fusion welding / non pressure welding



Arc welding process:

Arc welding is a group of welding process where in coalescence is produced by heating with an electric arc or arcs, mostly without the application of pressure and with or without the use of filler metal depending upon the base plate thickness. There are different arc welding process of which the one in workshop is flux shielded metal arc welding.

Flux shielded metal arc welding:

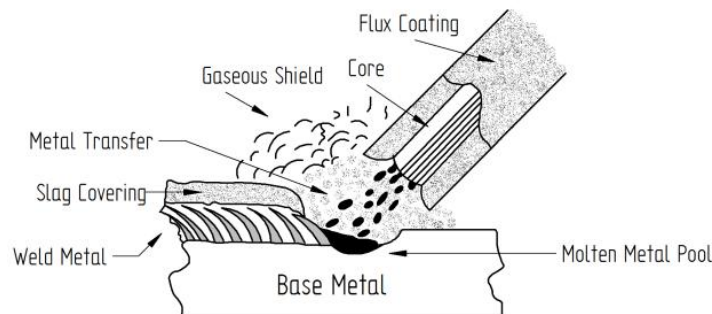
Definition: It is an arc welding process where in coalescence by heating the work piece with an electric arc setup between a flux coated electrode and the work piece. The flux covering

decomposes due to arc heat and performs many function, like arc stability, weld metal protection etc. the electrode itself melts and supplies the necessary filler metal.

Principle of the process:

Heat required for welding is obtained from the arc stuck between a coated electrode and the work piece. The arc temperature produced by the heat is about 4000 deg C and thus the arc heat can be increased or decreased by employing higher or lower arc currents. A high current with a smaller arc length produces a very intense heat. Both DC and AC may be used. For current over 750 amperes A.C equipment is preferred as it has high efficiency negligible loss at peak load and minimum maintenance.

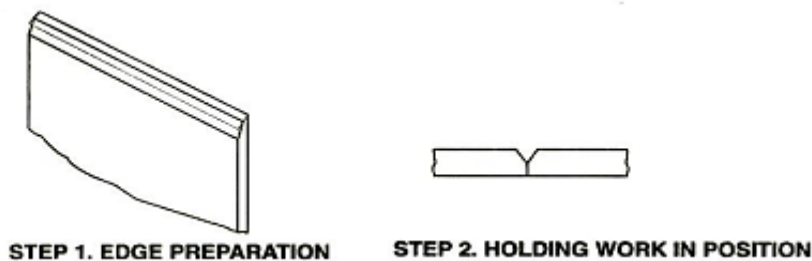
The arc melts the electrode end and the job. Material droplets are transferred from the electrode to the job, through the arc, and are deposited along the joint to be welded. The flux coating melts produces a gaseous shield and slag to prevent atmospheric contamination of the molten weld metal.



Arc welding process

Various types of edge preparation can be used, and the choice of the most suitable is influenced by a number of factors. Some of these, not necessarily in order of importance, are

- Type of process
- Type of work
- Position of welding
- Access for arc and electrode
- Volume of deposited weld metal
- Dilution
- Cost of preparing edges
- Shrinkage and distortion



Arc welding Procedure:

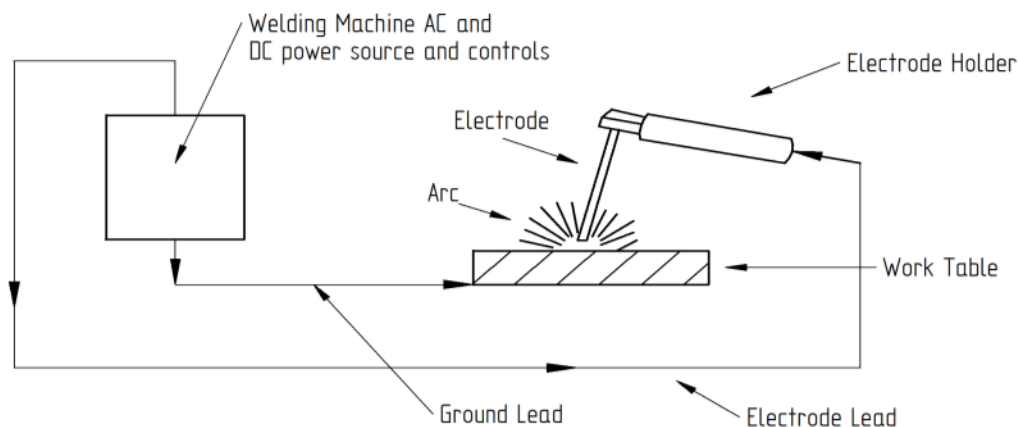
- The surface to be welded is cleaned and the edges of the plates may be filed for perfect joint and for more strength.
- The welding electrode is held in an electrode holder and the ground clamp is clamped to the surface plate and the work piece is placed on it for welding.
- The plates to be welded are positioned overlapping and tag weld is done on the ends to avoid the movement from one end of the plates.
- Now start welding from one end of the plates. The electric arc is produced melts the electrode and joins the two metal plates.
- Maintaining a gap of 3mm between the plates and the electrode for proper arc length. Complete the welding process by removing slag using wire brush and chipping hammer.

Tools and safety equipments in welding process

- Ac or Dc machines
- Electrode
- Electrode holder
- Cable and cable connectors
- Chipping hammer
- Earthing clamps
- Wire brush
- Helmet
- Safety goggles
- Hand gloves
- Apron
- Tongs
- Anvil etc.

Arc welding machines:

Both Alternate current and direct current are used for electric arc welding, each having its particular application.



The most commonly used power source for welding is AC supply. The equipment used is a Transformer, which changes high voltage, low amperage power to low voltage, high amperage welding power. The transformer used can be connected to single, two and three phase connection. They are either air-cooled or oil cooled depending upon their ampere rating. Generally current is in the range of 150 – 400 Amps with open circuit voltage varying from 30 to 100 volts.

Advantages of AC Arc welding

1. Fewer moving parts .no wear and tear
2. Little maintenance
3. The welding transformer and its controller are very much cheaper than the d.c. set of the same capacity
4. Suitable for ferrous metals, especially heavy steel sections.

Disadvantages

1. covered electrode must be used The a.c arc cannot be used satisfactorily for bare wire or light rods as with the d.c .arc
2. AC. uses higher voltage than d.c. shock risk is greater with AC
3. welding of cast iron ,bronze, and aluminum cannot be done as successfully as with d.c

Current requirements

20-200 amperes for thin materials

20-300 amperes for general work

Up to 600 amperes for heavy work.

Welding Electrodes

The types of electrodes that are generally used for welding are

- **Consumable Electrode** – it melts and supplies filler metal to the weld
- **Non – Consumable Electrode** - it does not melt but separate filler metal is to be added to the weld.
- **Bare electrodes:** which consist of metal or alloy wire without any flux coating.
- **Flux covered (coated) electrodes:** which have flux covering the metal or alloy. The flux acts as a shield to protect the arc and also to protect the weld pool.



Specification of electrode:

The coated electrodes are further classified in to light coated and heavy coated. Heavy coated has layer of 1mm to 3 mm of coating, Heavy coatings are composed of ionizing (chalk),

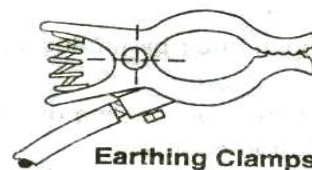
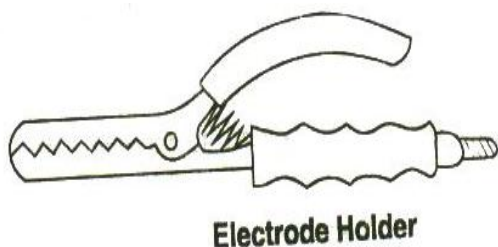
deoxidizing (aluminum, Ferro magnetic et.), gas generating (starch), slag forming (kaolin), alloying and binding material. For hand arc welding the electrodes has diameter up to 12 mm and 450 mm long are used.

Coated electrode has several purposes:

- To facilitate the establishment and maintenance of arc
- To protect the molten metal from oxygen and nitrogen of air by producing gas shielded around the arc and weld pool.
- To assist in the formation of slag so as to prevent the rapid cooling of the weld seam.
- To provide the alloying element that is not present in cold wire.

Electrode holder:

A metal electrode is the device used for holding the electrode mechanically. It conveys electric current from welding machine to the electrode; it has an insulated handle to protect the operator's from heat.



Chipping hammer:

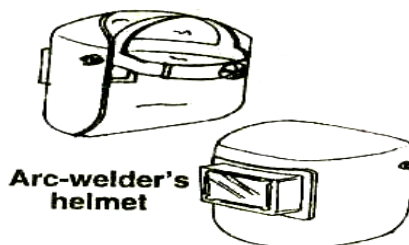
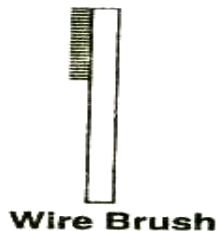
Chipping hammer is used for chipping the slag coating on weld surface. It consists of a long handle with head made of cast steel.

Earthing Clamps:

It is connected to the end of the ground cable. It is normally clamped to the welding table or the job itself to complete the electric current.

Wire brush:

The wire brush is used to clean the surface to be welded.



Helmet:

It is known as face shield or helmet. It is used to protect the eyes of the welders from the light sparks produced during welding. It is normally held in hand.

Safety goggles:

Goggles with glasses are used to protect the eyes of the welder from the light sparks produced during welding.



Safety Goggles



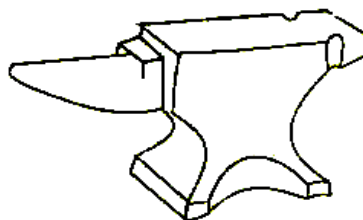
Flat Tongs

Flat tongs:

When welding smaller pieces together, for moving the pieces to different welding angles and shifting of pieces for chipping. It is advisable to use a flat tongs.

Anvil:

It is a solid casting of mild steel. Used for supporting the work for hammering, punch marking, chipping of welded parts etc.



Anvil

WELDING DEFECTS:

- Cracks
- Distortion
- Incomplete penetration
- Inclusions
- Porosity and Blow holes
- Poor fusion

- Spatter
- Under cutting

Gas Welding

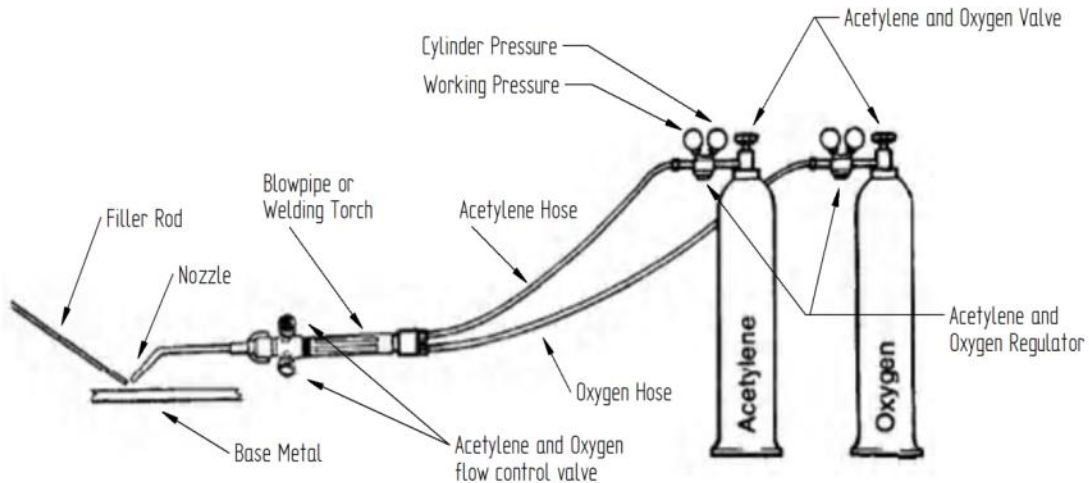
- Oldest methods of fusion welding process
- Most widely used method of metal melting
- The equipment is relatively simple and cheap
- Heat is generated by the combustion of combustible gas with oxygen
- Combustion takes place at the nozzle or the outlet of the torch
- This process of welding is generally known as Oxygen – Fuel Gas welding (OFG)
- Commercial gases used for gas welding: acetylene, hydrogen, propane, butane and commercial LPG.

Oxy – Acetylene Gas welding

- Common gas welding process.
- Acetylene is the fuel gas used, produces high heat content in the range of 3200°C
- Acetylene gas has more available carbon (92.3%) and hydrogen (7.7%) by weight.
- Heat is released when the carbon breaks away from hydrogen to combine with O₂ and burn.



GAS WELDING EQUIPMENT



Advantages of gas welding

1. The equipment is inexpensive, uncomplicated.
2. It is portable.
3. Useful for welding light metals and for repair jobs.
4. Gas welding can be used with all the common metals.

Disadvantages

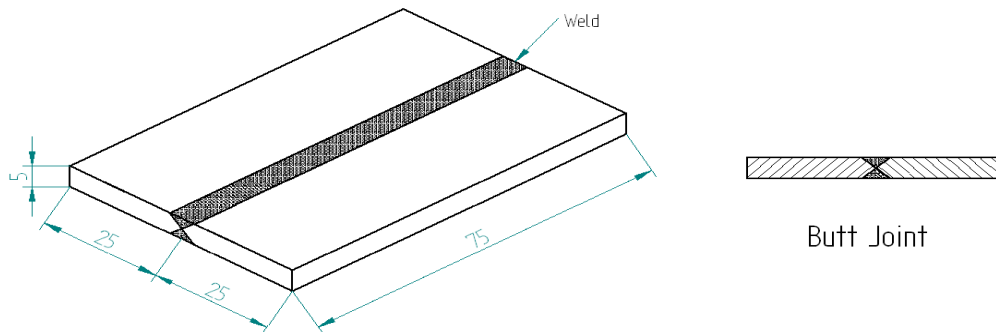
1. Acetylene is explosive.
2. Gas welding is slower than electric arc welding.
3. Heated areas are larger and cause more distortions.
4. The process is not satisfactory for heavy section.

Experiment Models

Model 1: Butt joint

Aim: To join two given metal plates to obtain a **Butt joint** by arc welding process

Tools Required: Welding transformer, welding electrodes, safety gloves, hand shield, chipping hammer, wire brush etc



(All dimensions in mm)

Procedure:

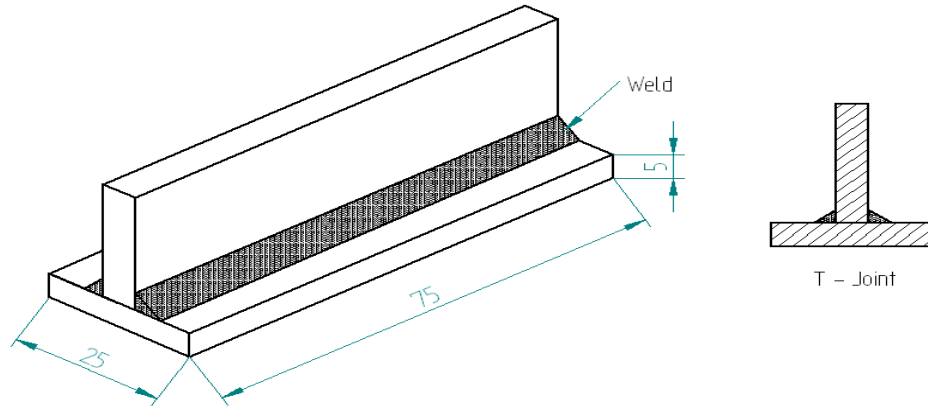
- The surface to be welded is cleaned and the edges of the plates are grinded in such a way that it forms a V and an inverted V shape when the plates butt each other.
- The welding electrode is held in an electrode holder and the ground clamp is clamped to the surface plate and the work piece is placed on it for welding.
- The plates to be welded are positioned overlapping and tag weld is done on the ends to avoid the movement from one end of the plates.
- Now start welding from one end of the plates.
- The electric arc is produced melts the electrode and joins the two metal plates.
- Maintaining a gap of 3mm between the plates and the electrode for proper arc length.
- Complete the welding process by removing slag using wire brush and chipping hammer.

Result: Thus the two metal plates are welded to form a double V butt joint.

Model 2: 'T' joint

Aim: To join two given metal plates to obtain a '**T**' joint by arc welding process

Tools Required: Welding transformer, welding electrodes, safety gloves, hand shield, chipping hammer, wire brush etc.



(All dimensions in mm)

Procedure:

- The surface to be welded is cleaned and the edges of the plates may be filed for perfect joint and for more strength.
- The welding electrode is held in an electrode holder and the ground clamp is clamped to the surface plate and the work piece is placed on it for welding.
- The plates to be welded are positioned overlapping and tag weld is done on the ends to avoid the movement from one end of the plates.
- Now start welding from one end of the plates.
- The electric arc is produced melts the electrode and joins the two metal plates.
- Maintaining a gap of 3mm between the plates and the electrode for proper arc length.
- Complete the welding process by removing slag using wire brush and chipping hammer.

Result: Thus the two metal plates are welded to form a 'T' joint.

SHEET METAL (Development & Preparation of the models)

Introduction to sheet metal work:

Sheet metal work is working with metal sheets from 16 – 30 gauge, using hand tools and simple machines. By marking, development, cutting, forming in to shape and joining to fabricate many domestic utility items, machine covers, hoppers, guards, tanks, stacks, duct work, pipes, bend, boxes, etc. common metals used in sheet metal work are black iron, galvanized iron, stainless steel, copper, brass, zinc, aluminum, tin plate and lead.

The material, which is used in workshop, is galvanized iron. It is zinc coated hence it is known as galvanized iron. This sheet can withstand contact with water and exposure to atmosphere.

Development: It is method of laying out surfaces of the pattern with suitable allowances in full size on metal sheet.

Methods of development: There are four methods of development

- 1) **Parallel line development :** This is used to develop cubes , prisms, cylinders, and similar objects
- 2) **Radial line development :** This is used to develop pyramids, cones and similar objects
- 3) **Triangular development :** This is used to develop transition piece, pipes and objects of various shapes
- 4) **Approximate Development:** This is to develop sphere, ellipsoid etc.

Sheet metal operations:

- 1) **Shearing:** shearing is a cut in straight line across the strip, sheet or bar. This procedure leaves a clean edge on the metal that is sheared or cut
- 2) **Punching :** punching is the operation of producing circular holes on sheet metal by punch or die
- 3) **Drawing:** this is operation of producing thin walled hollow or vessel shaped parts from sheet metal.
- 4) **Notching:** this is process of removal metal to the desired shape from the side or edge of sheet or strip to prevent overlapping and bulging of seams, hems and edges.
- 5) **Flattering:** It is the operation in which the sheet is made flat by heating the metal down where it stands up from the sheet.

Sheet metal joints: Hem and seams

Sheet metal working incorporates a wide variety of hems and seams. A **hem** is an edge made by folding. It stiffens the sheet metal and does away with the sharp edge.

A **seam** is a joint made by fastening two edges together.

Gauge: gauge is only a number to know the thickness of sheet and diameter of wire.

GAUGE number	Thickness or diameter
10 G	3.25mm
12G	2.64mm
14 G	2.03mm
16 G	1.63mm
18 G	1.22mm
20 G	.91mm
22 G	.71mm
24 G	.56mm
26 G	.46mm
28 G	.38mm
30 G	.30mm

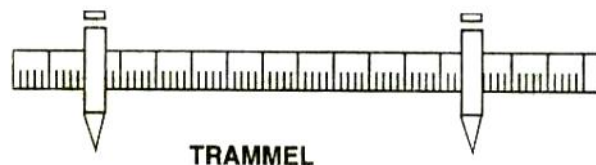
Tools used in sheet metal work:

1. Ball peen hammer
2. Snip
3. Straight edge
4. Stake
5. Rivets
6. Steel rule
7. Scriber
8. Mallet
9. Trammel etc.

Some of these are mentioned in fitting section.

Trammel point:

It is used to draw large circles and arcs. The trammel has two removable pointed legs and mounted on a separate holder.



Cutting tools:

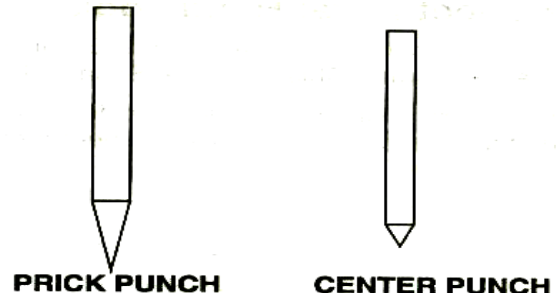
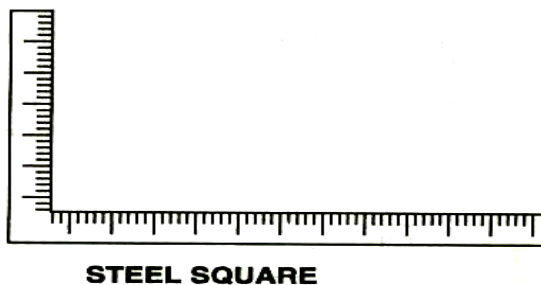
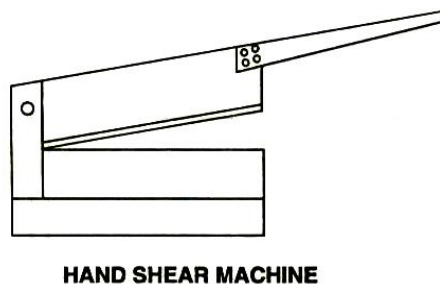
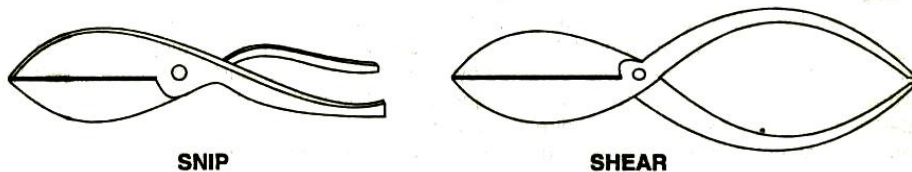
Snips are used to shear or cut the metal sheets to the sheets to the required size and shape. Snips are used to cut thin sheet metal. The following snips metals are used in sheet metal work.

Straight snip:

It is used to cut or trim along a straight line. The blades in this snip are straight.

Bend snip:

It is used to trim or cut along inside curves. The blades in this snip are curved back from the cutting edge, which permits the sheet to slide over the top blades while cutting.



Prick punch:

It is a sharply pointed tool. The tapered point of the punch has an angle of 40° . It is used to make small punch marks on layout lines in order to make them last longer.

Center punch:

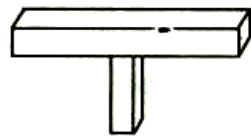
Looks like a prick punch. Its point has an angle more obtuse than that of a prick punch. The angle is around 60° .

Stakes:

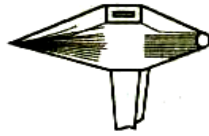
The stakes are the supporting tools in sheet metal work to form a shape. And it is used when the sheet metal bending, riveting, punching etc.

Hand stake:

It is used for pressing the inner sides of a straight joint in the sheet. It has a flat surface with two straight edges, a concave and a convex edge.



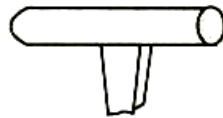
Hatchet



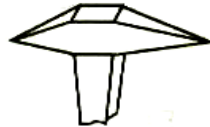
Funnel



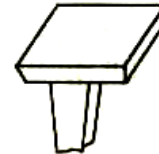
Bottom



Tubular



Beakhorn



Square

STAKES

Half round stake:

It is used to form a round seam joint on the inner side of the job.

Horse stake:

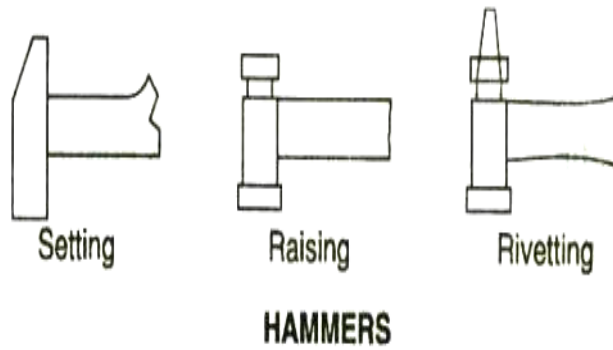
It has two square holes for holding one or two stakes to carry out different operations on the job.

Taper stake:

It is used to form a conical or tapering job.

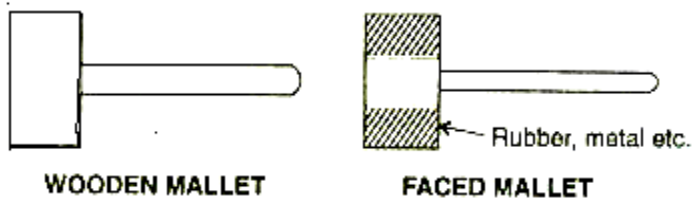
Hammers:

Hammers are used for forming shapes by hollowing, stretching or raising. There are many types of hammers in use in sheet metal shop. (Some other hammer mentioned in fitting).



Mallets:

Mallets are soft hammers and are made of raw hide, hard rubber, copper, brass, Lead, or wood. It is mostly used to strike soft metal and give light blow on sheet metal.



SOLDERING

- A joining process wherein coalescence is produced by heating to a suitable temperature and by using a filler metal having a melting point not exceeding 427°C and below the solidification temperature of the base metals”
- The filler metal fills in the gap of the joint by capillary action.
- Soldering uses fusible alloys to join metals known as solder
- Ordinary gas flames or electric soldering iron is used to supply the heat to melt the solder.
- Fluxes are used with solder in soldering process.
- Fluxes are defined as any solid, liquid or gaseous material when heated accelerates the wetting of metal with the solder
- Due to wetting molten solder flows into the joint and fills the space between the two pieces to be soldered
- At elevated temperature flux is highly reducing in nature preventing the formation of metal oxides
- Fluxes that are generally used in soldering are Rosin, Zinc Chloride and Aluminum Chloride

Soldering iron:

Soldering iron is used for soldering work and it consists of a copper bit held by a steel rod and wooden handle. There are many types of soldering iron like copper bit, hatchet adjustable and soldering iron.



COPPER BOX SOLDERING IRON



ELECTRIC SOLDERING IRON

Composition of solder:

Usually tin 60% and lead 40% used for electrical work.

Tin 50% and lead 50% used machine soldering.

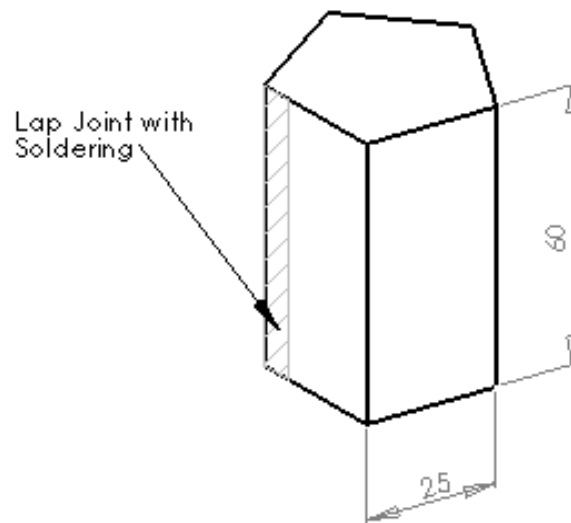
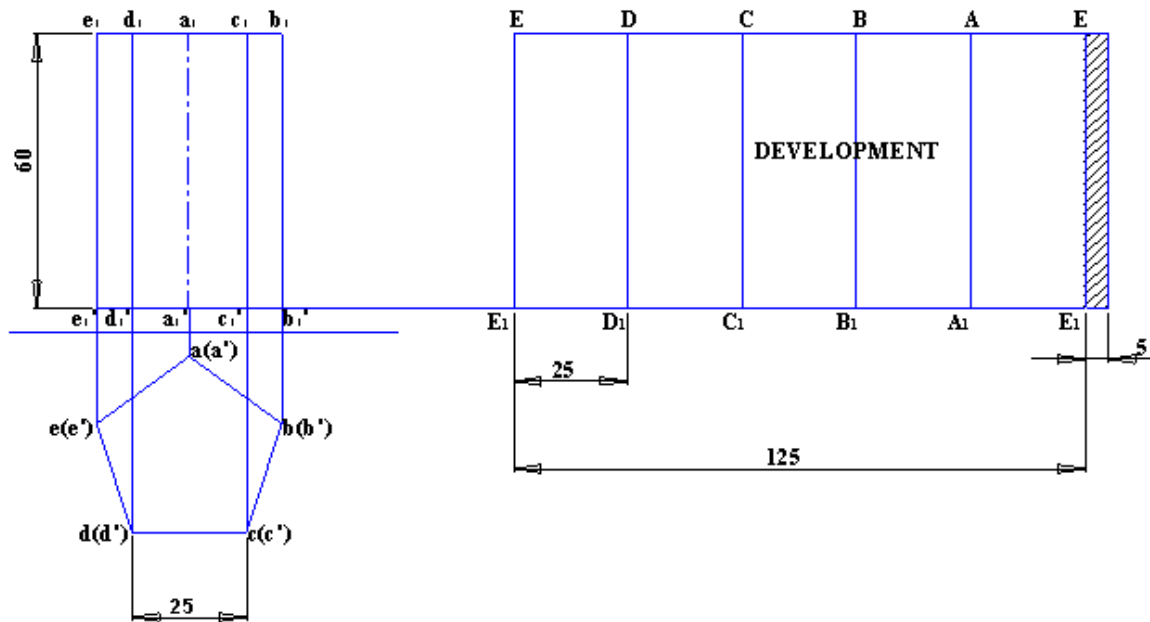
Tin 30% and lead –70% used for plumber solder.

Brazing:

Brazing is a joining process, which produces coalescence of materials by heating to a suitable temperature and using a filler metal having a melting temperature above 427°C and below the solidification temperature of the base metals being joined. The filler metal is distributed between the closely fitted surfaces of the joint by capillary attraction. Brazing is distinguished from soldering in that soldering employs a filler metal having a melting point below 427°C .

EXPERIMENTAL MODELS

Model No. 1: Pentagonal Prism

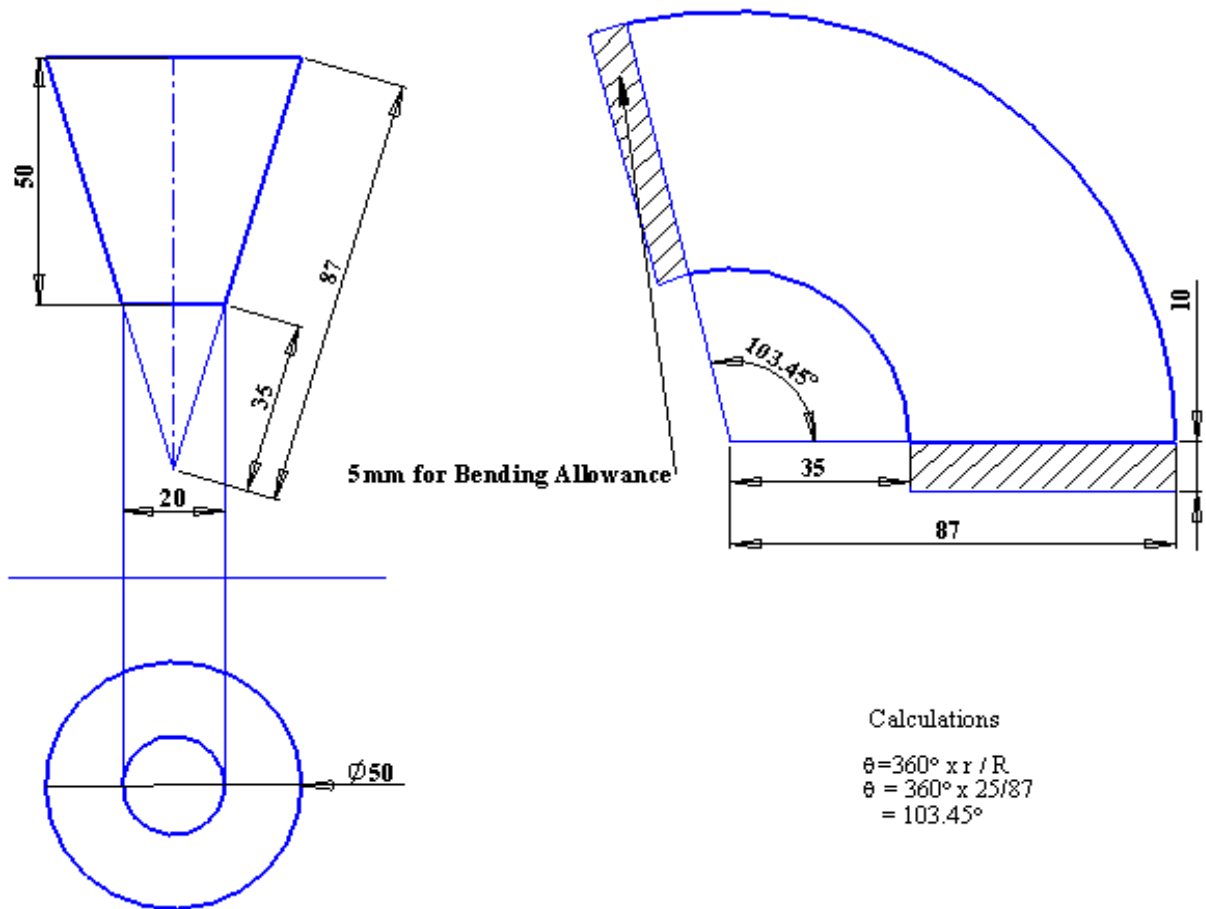


Isometric View

PROCEDURE FOR PREPARING PENTAGONAL PRISM

1. Draw the orthographic view as shown in above figure
2. Layout the five rectangular vertical faces of pentagonal prism
3. Set 5mm extra allowance for joining at the ends.
4. Trace the development on given G.P.sheet (galvanized plain sheet 28 gauge) and mark all bending lines
5. Cut the sheet along the line according to the development shape
6. Bend the seam line using bending dies
7. Finish all rectangular faces and longer edges using rectangular faced stakes
8. Dress by mallet and join the ends
9. Solder the end joint using electric soldering

Model No 2: Frustum of Cone



PROCEDURE FOR PREPARING A FRUSTUM OF CONE

1. Draw the front view of of right circular cone OAB of base dia .is 50mm.A section plane cuts perpendicular to the axis of cone at 50mm height
2. With “O” as center radius equal to slant generator length (OA or OB) draw an arc. With same center “O” radius equal to (OC or OD) draw another arc.

3. Find
$$\theta = \frac{360 \times r}{R}$$

Where

r= radius of base circle of cone

R= Length of slant generator of cone

θ = Angle subtended to cut arc.

4. Set an angle at point of vertex it cut arc at the points EFGH .set off 5 and 5+5 mm extra for seam joint
5. Trace the development part on given G.P. sheet. Mark all necessary lines.
6. Cut the sheet along line according to shape of development.
- 7Fold extra allowances in clockwise and anticlockwise directions by keeping hacksaw blade thickness and pressed
8. Remove hacksaw blade, and bend main body using cone stake and lock end joint and lock

ENGINE LATHE

A **lathe** is a machine tool which rotates the work piece on its axis to perform various operations such as cutting, sanding, knurling, drilling, or deformation with tools that are applied to the work piece to create an object which has symmetry about an axis of rotation.

Lathes are used in woodturning, metalworking, metal spinning, and glass working. Lathes can be used to shape pottery, the best-known design being the potter's wheel. Most suitably equipped metalworking lathes can also be used to produce most solids of revolution, plane surfaces and screw threads or helices. Ornamental lathes can produce three-dimensional solids of incredible complexity. The material can be held in place by either one or two *centers*, at least one of which can be moved horizontally to accommodate varying material lengths. Other work holding methods include clamping the work about the axis of rotation using a chuck or collet, or to a faceplate, using clamps or dogs.

Examples of objects that can be produced on a lathe include candlestick holders, cue sticks, table legs, bowls, baseball bats, musical instruments (especially woodwind instruments), crankshafts and camshafts. The parts of the Lathe are illustrated in Figure 1.

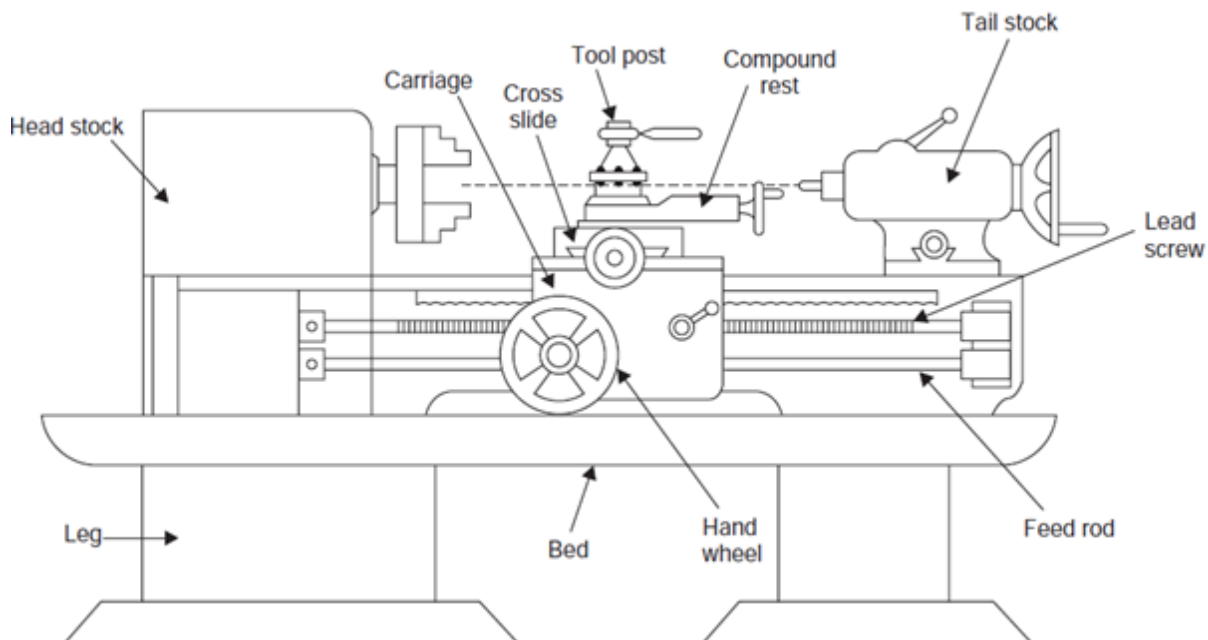


Figure 1: Parts of the Lathe

The Bed: The lathe bed is a mounting and aligning surface for the other machine components. Viewed from the operating position in front of the machine, the headstock is mounted on the left end of the bed and the tailstock on the right. The bed must be bolted to a base to provide a rigid and stable platform. The bed ways are a precision surface (or surfaces) on which the carriage slides left and right during machining operations. The ways are machined straight and flat and are either bolted to the top of the bed or are an integrally machined part of the bed.

Headstock: The headstock holds the spindle and drive mechanism for turning the work piece. The spindle is a precision shaft and bearing arrangement rotated directly by a motor or through a motor-driven belt. Gears or sliding pulleys mounted at the rear of the headstock allow spindle speed adjustment. A work piece is held in the spindle for turning or drilling by a jawed chuck or a spring collet system. Large, unusual shaped, or otherwise difficult to hold pieces, can be attached to the spindle with a face plate, drive dogs and special clamps.

Tailstock: The tailstock supports long work that would otherwise sag or flex too much to allow for accurate machining. Without a tailstock, long pieces cannot be turned straight and will invariably have a taper. Some tailstocks can be intentionally misaligned to accurately cut a taper if needed. The tailstock has a centering device pressed into a shallow, specially drilled hole in the end of the work piece. The center can be either "live" or "dead." Live centers have a bearing, allowing the center to rotate along with the work piece. Dead centers do not rotate and must be lubricated to prevent overheating due to friction with the work piece. Instead of a center, a drill chuck can be mounted in the tailstock.

Carriage: The carriage provides mounting and motion control components for tooling. The carriage moves left and right, either through manual operation of a hand wheel, or it can be driven by a lead screw. At the base of a carriage is a saddle that mates and aligns with the bed ways. The cross-slide, compound rest and tool holder are mounted to the top of the carriage. Some carriages are equipped with a rotating turret to allow a variety of tools to be used in succession for multi-step operations.

Cross Slide: The cross-slide is mounted to the top of the carriage to provide movement perpendicular to the length of the bed for facing cuts. An additional motion assembly, the compound rest, with an adjustable angle, is often added to the top of the cross slide for angular

cuts. The cutting tools that do the actual metal removal during turning are mounted in an adjustable tool holder clamped to the compound rest.

Lead Screw: The lead screw provides automatic feed and makes thread cutting possible. It is a precision-threaded shaft, driven by gears as the headstock turns. It passes through the front of the carriage apron and is supported at the tailstock end by a bearing bracket. Controls in the apron engage a lead nut to drive the carriage as the lead screw turns.

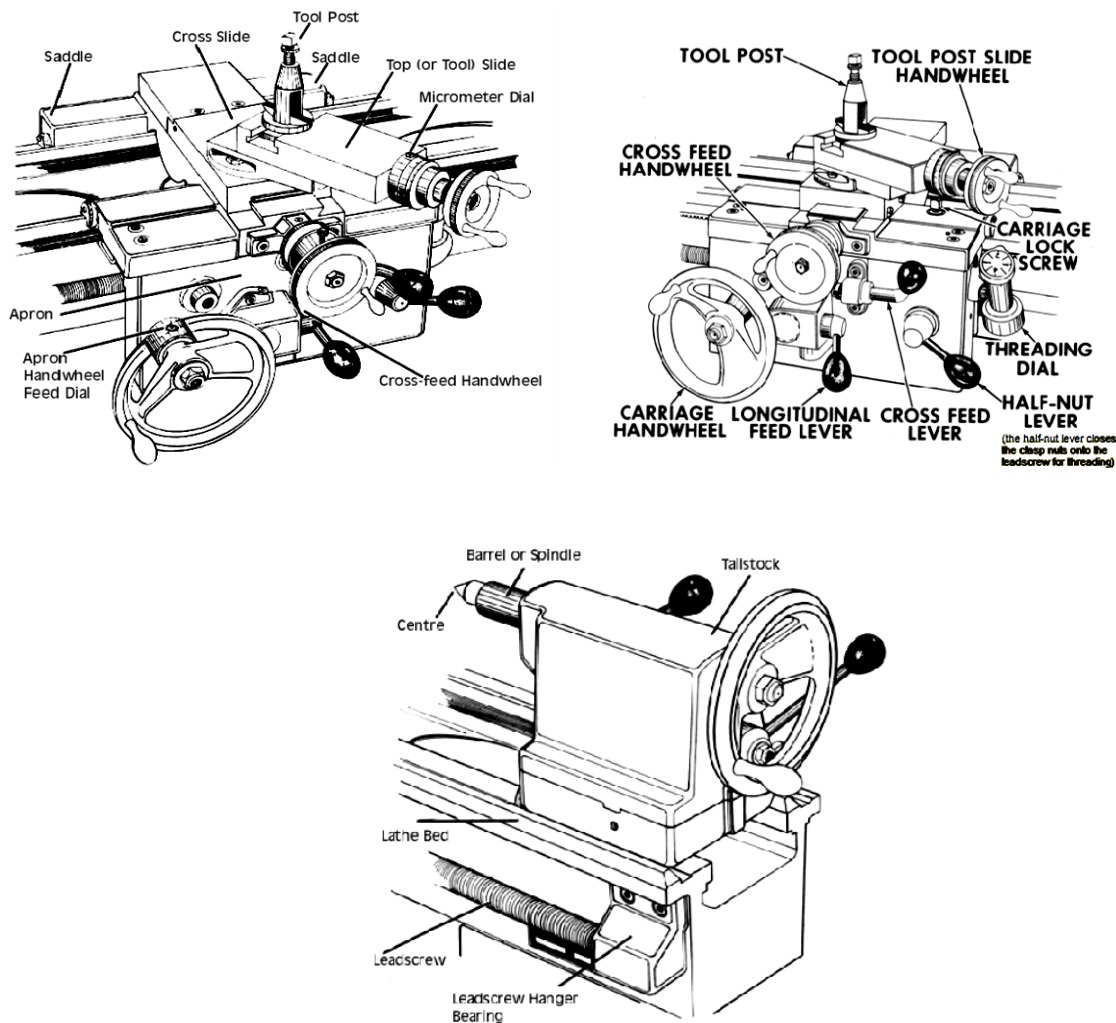


Figure 2: Parts of the Lathe

Specifications of the Lathe

1. Swing over machine bed
2. Swing over cross slide
3. Distance between centers

4. Turning Length
5. Width of bed
6. Work Spindle diameter in front bearing
7. Spindle Bore
8. Taper bore according to DIN 228(shortened)
9. Normal Chuck diameter
10. Max. Diameter of face plate and clamping disk
11. Tool Slide & Guide Length of carriage
- 12 Graduation of Longitudinal scale on apron box hand wheel
- 13 Cross-slide travel
- 14 Graduation of scale at cross slide spindle
- 15 Width of cross-slide
- 16 Compound slide travel
- 17 Graduation of scale on compound slide spindle
- 18 Width of top slide
- 19 Tailstock with hand wheel
- 20 Diameter of center sleeve

Types of Lathe

Engine Lathe

The most common form of lathe, motor driven and comes in large variety of sizes and shapes.

Bench Lathe

A bench top model usually of low power used to make precision machine small work pieces.

Tracer Lathe

Lathe that has the ability to follow a template to copy a shape or contour.

Automatic Lathe

Lathe in which the work piece is automatically fed and removed without using an operator. Cutting operations are automatically controlled by a sequencer of some form

Turret Lathe

lathe which have multiple tools mounted on turret either attached to the tailstock or the cross-slide, which allows for quick changes in tooling and cutting operations.

Computer Controlled Lathe

A highly-automated lathe, where both cutting, loading, tool changing, and part unloading are automatically controlled by computer coding.

CONVENTIONAL TYPES OF LATHE

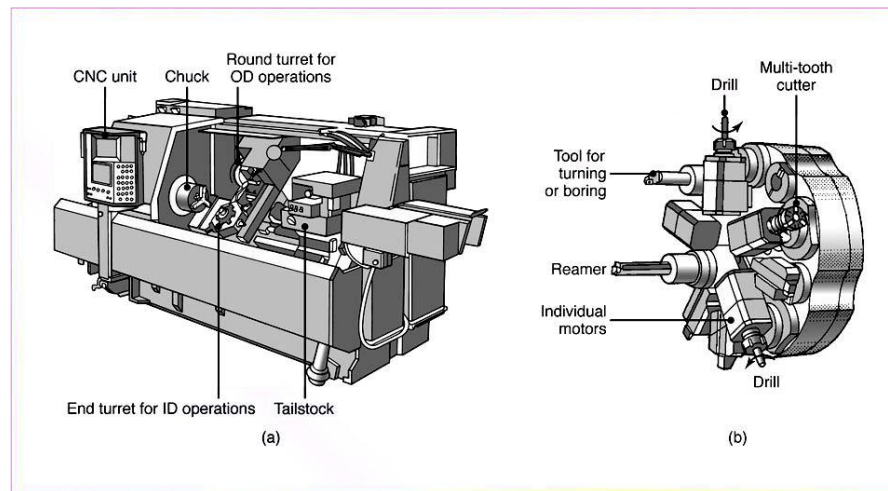


Figure 3: Conventional type of Lathe

Lathe Operations

Turning: produce straight, conical, curved, or grooved work pieces

Facing: to produce a flat surface at the end of the part or for making face grooves.

Boring: to enlarge a hole or cylindrical cavity made by a previous process or to produce circular internal grooves.

Drilling: to produce a hole by fixing a drill in the tailstock

Threading: to produce external or internal threads

Knurling: to produce a regularly shaped roughness on cylindrical surfaces

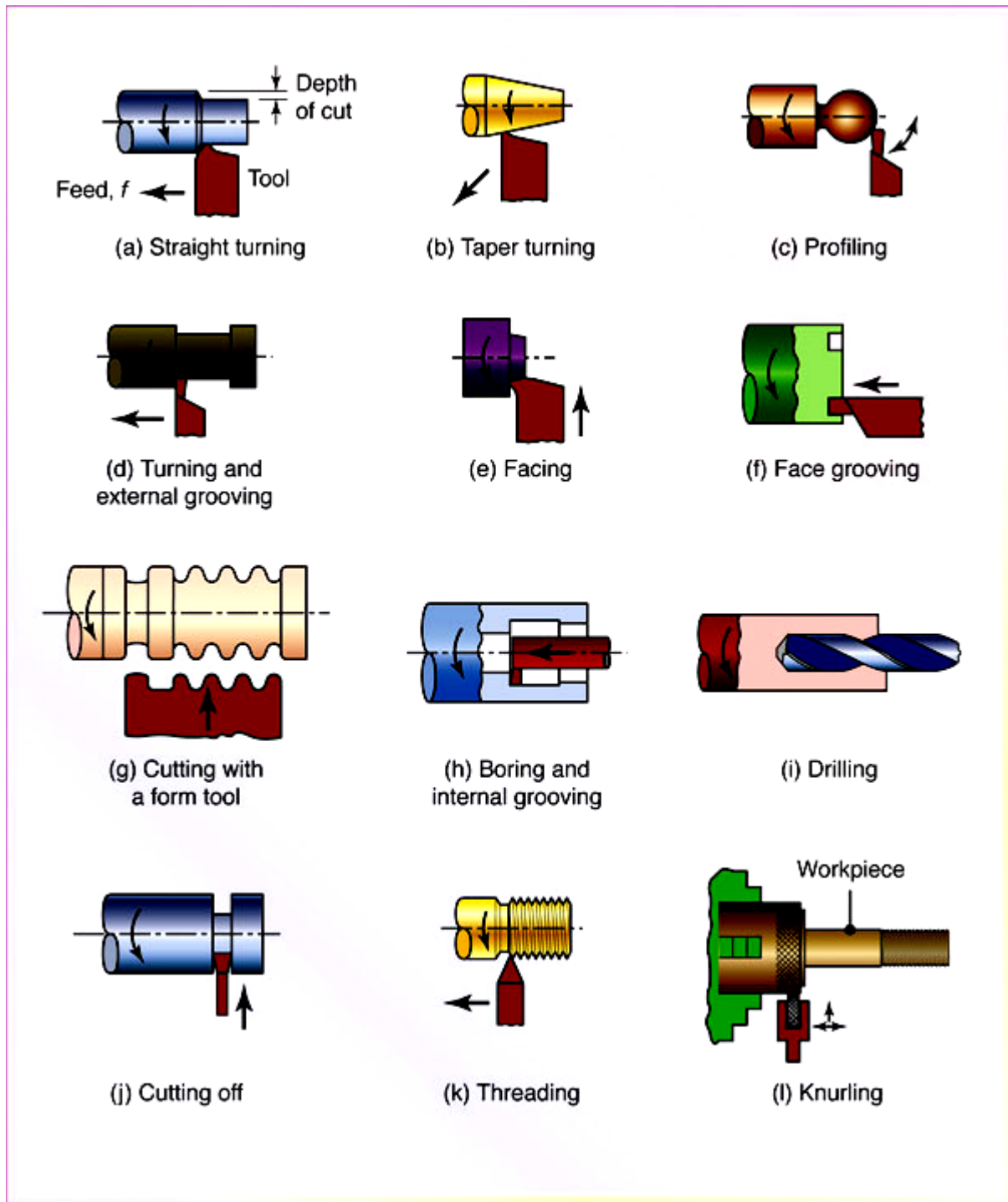


Figure 4: Lathe operations

Work Holding Devices

Three jaw chuck: For holding cylindrical stock centered and for facing/center drilling the end of your aluminum stock.

Four-Jaw Chuck: - This is independent chuck generally has four jaws , which are adjusted individually on the chuck face by means of adjusting screws



Figure 5(a): Three Jaw Chuck

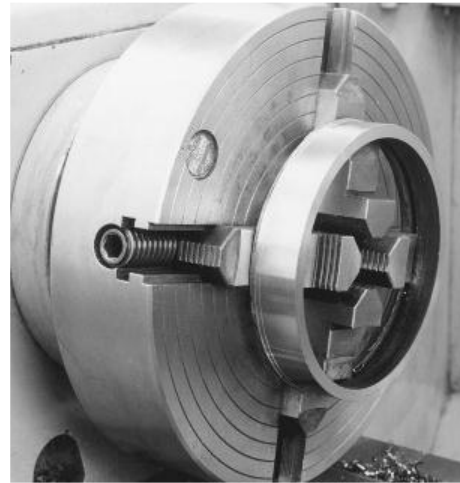


Figure 5(b): Four Jaw Chuck

Collet Chuck: Collet chuck is used to hold small workpieces

Magnetic Chuck: Thin jobs can be held by means of magnetic chucks.



Figure 6: Collet Chuck



Figure 7: Magnetic Chuck

GENERAL PURPOSE CUTTING TOOLS

The lathe cutting tool or tool bit must be made of the correct material and ground to the correct angles to machine a workpiece efficiently. The most common tool bit is the general all-purpose bit made of high-speed steel. These tool bits are generally inexpensive, easy to grind on a bench or pedestal grinder, take lots of abuse and wear, and are strong enough for all-around repair and fabrication. High-speed steel tool bits can handle the high heat that is generated during cutting and are not changed after cooling. These tool bits are used for turning, facing, boring and other lathe operations. Tool bits made from special materials such as Single point tool bits can be one end of

a high-speed steel tool bit or one edge of a carbide or ceramic cutting tool or insert. Basically, a single point cutter bit is a tool that has only one cutting action proceeding at a time. A machinist or machine operator should know the various terms applied to the single point tool bit to properly identify and grind different tool bits

SINGLE POINT TOOL BITS

The *shank* is the main body of the tool bit. The *nose* is the part of the tool bit which is shaped to a carbides, ceramics, diamonds, cast alloys are able to machine workplaces at very high speeds but are brittle and expensive point and forms the corner between the side cutting edge

For normal lathe work, high-speed steel tool bits are used. The *nose radius* is the rounded end of the tool bit.

The *face* is the top surface of the tool bit upon which the chips slide as they separate from the work piece. The *side or flank* of the tool bit is the surface just below and adjacent to the cutting edge. The *cutting edge* is the part of the tool bit that actually cuts into the workpiece, located behind the nose and adjacent to the side and face. The *base* is the bottom surface of the tool bit, which usually is ground flat during tool bit manufacturing. The end of the tool bit is the near-vertical surface which, with the side of the bit, forms the profile of the bit. The end is the trailing surface of the tool bit when cutting. The *heel* is the portion of the tool bit base immediately below and supporting the face.

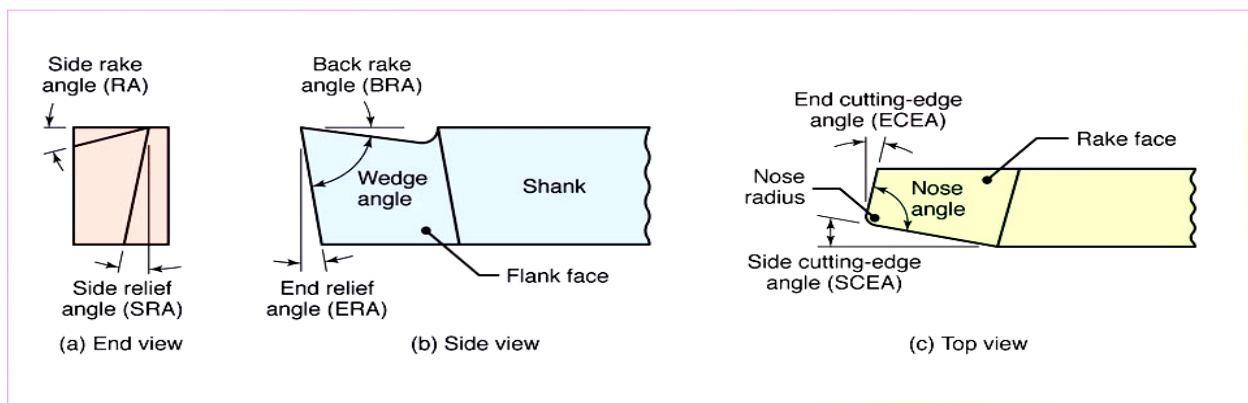


Figure 8: Nomenclature of single point cutting tool

MACHINING

Machining is a manufacturing process in which a cutting tool is used to remove excess material from a workpiece. The material that remains is the desired part geometry. The cutting tool deforms the workpiece in shear and creates scrap called “chips.” As chips fall off the workpiece a new

surface is exposed. Almost all solid metals, plastics, and composites can be machined by conventional machining. Machining can create any regular geometry, i.e., planes, round holes, and cylinders. Machining can produce dimensions to tolerances of less than 0.001" (0.025mm) Surface finishes of better than 16 μ in (0.4 μ m) can be produced by machining processes. A cutting tool has one cutting edge (facing tool or turning tool) or more than one cutting edges (drill, end mill). The cutting edge separates the chip from the workpiece.

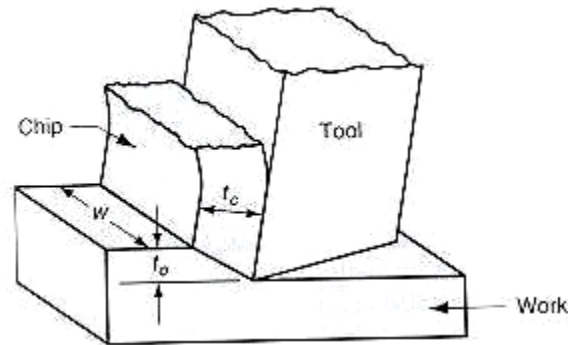


Figure 9: Formation of chip

The rake face of a tool guides the chip from the surface of the workpiece and is oriented at an angle α . The rake angle α is measured relative to a plane perpendicular to the work surface. The flank of a tool provides clearance between the cutting tool and the newly exposed surface to protect the surface from abrasion. The flank is oriented at an angle called the relief angle. The picture below illustrates the make-up of a cutting tool.

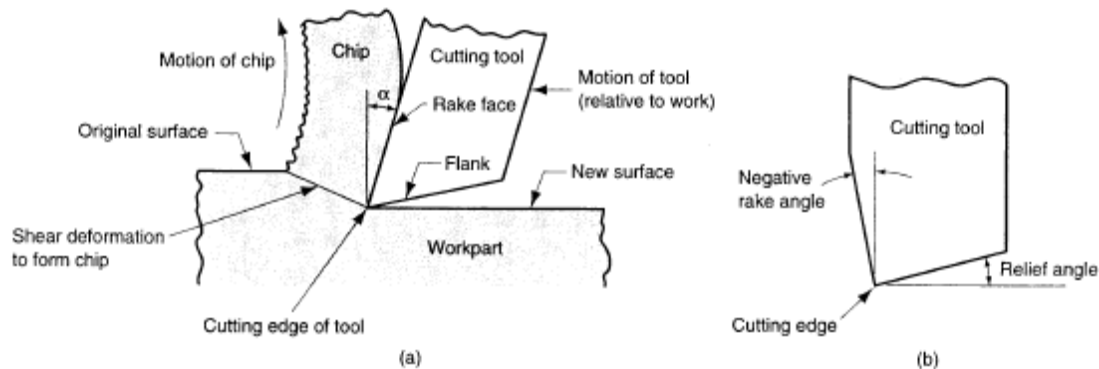


Figure 10: Nomenclature of tool-workpiece interphase

Discontinuous chips: This type of chip is usually formed when cutting hard, brittle materials, partly because these materials cannot withstand high shear forces and therefore the chips formed shear cleanly away. However, the chips formed may be firmly or loosely attached to each other or may leave the cutting area in a fine shower – as often encountered when cutting hard Brass. When

discontinuous chips are formed, there is a greater possibility of tool chatter; unless the tool, tool-holder and workpiece are held very rigidly; due to pressure at the tool tip increasing during chip formation and then releasing suddenly as the chip shears.

Continuous chips: This type of chip is usually formed when cutting soft or ductile materials such as Aluminium or Copper. There is less likelihood of chatter and surface finish is usually better than when discontinuous chips are formed. A disadvantage of continuous chips is the fact that they can become very long and become entangled with the machine or pose a safety hazard. This problem can be overcome by the use of chip-breakers; a device clamped to the top of the tool that encourages the chip to curl more tightly, hitting the workpiece and breaking off.



Figure 11(a): Discontinuous chips

Figure 11(b): Continuous chips

Built Up Edge: Figure shows a tool with a Built-up Edge (B.U.E.). A B.U.E. is formed when particles of the workpiece material weld to the rake face of the tool during cutting. Large B.U.E.s can be very detrimental to surface finish and integrity, they effectively change the geometry of the cutting edge and consequently shear plane angle, this can lead to residual stresses in the material below the depth of cut. As a large B.U.E. dislodges it can deposit work hardened particles, to become embedded in the finished surface. A thin, stable B.U.E. is generally considered desirable as this can tend to reduce frictional wear on the rake face of the tool.

Tool Wear: Wedge shape cutting tools normally wear in two ways. Figure shows the typical wear pattern of a wedge shape cutting tool. Crater wear occurs on the rake face of the tool just behind the cutting edge and is caused by the rubbing of the chip across the surface. Flank wear occurs on the clearance angle of the tool causing rubbing and degradation of the surface finish. Cutting tools are deemed to have failed and require regrinding or replacing when flank wear exceeds 0.25mm or when cratering appears, this allows regrinding with minimal removal of tool material. It can be seen from Figure that when cratering appears the cutting edge becomes thinned and less able to

dissipate the heat generated during cutting; leading to unpredictable and possibly catastrophic, failure.

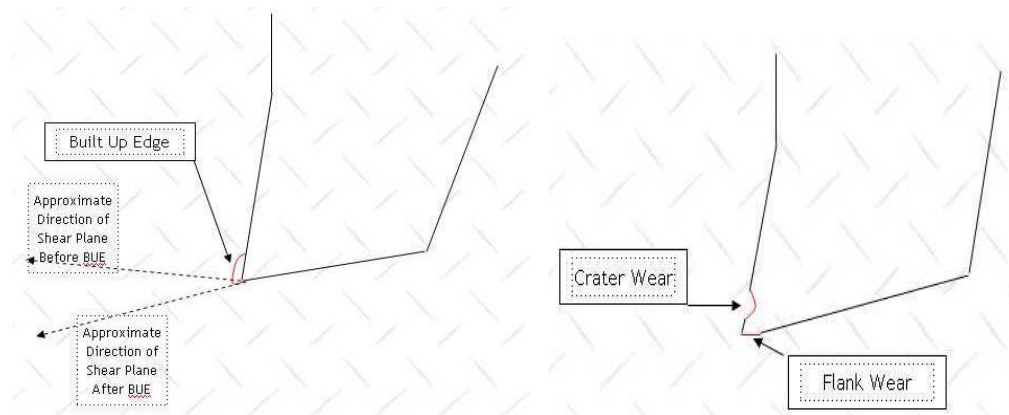


Figure 12: Phenomenon of Cratering

TURNING MODEL

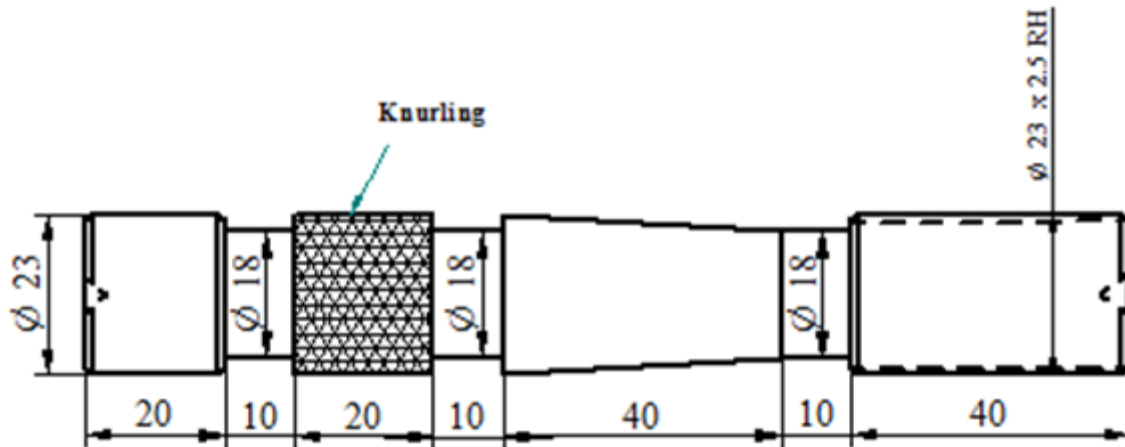


Figure: Model 1

Aim: To complete the given material to the specified dimension

Apparatus/Components required: Lathe with revolving center, outside calipers, center drill bit, single point cutting tool center, outside calipers, center drill bit.

Procedure:

1. Take the raw material and necessary tools from the stores.
2. Compare the model and the raw material. Identify the variation in dimension i.e. length and diameter of the model to specimen
3. Reduce the length of the specimen to 150mm,
 - a) Fix the specimen in a three-jaw chuck to reduce the length by 2 to 3mm by facing one side of the specimen.
 - b) Measure 150mm from the faced surface and mark it on the specimen.
 - c) Fix the work piece in reverse direction and face it with the marked length. Check for the reduced length of the model.
4. Reduce the diameter of specimen to 23 mm.
 - a) To hold the work piece maximum machinable exposing length.
 - b) To avoid wobbling the work pieces must be mounted between chuck and dead center. Therefore, a Center hole is to be drilled on the other end. To drill a center,

hole the work piece is held in a chuck center drill bit is mounted on tailstock sleeve, and drill is fed to job by rotating tailstock wheel, Plain turn the work piece to a diameter of 23 mm.

5. Step turn the work piece to 18 mm diameter.

- a) Mark the variation in diameter along the length.
- b) Mark the points of variation diameter along the length of the work piece.
- c) Step turn the work piece to the required diameter as shown in Figure 16.
- d) Step turn is done in such a way that chamfer at the step turn is 45 degrees. This is taken care by using a cutting tool whose cutting edges are at 45 degrees.

6. Knurling.

- a) Decide the type knurling required Choose the specific combination of knurling wheels on the knurling tool.
- b) Fix the knurling tool on the tool post where tool was generally being fixed. Engage the selected knurl wheels and engage them onto the workpiece. Turn the workpiece with the workpiece engaged with the knurling tool. Note: use the correct combination of the knurl wheels.
- c) Reduce the speed of spindle while thread cutting by using mechanism. The bull gear pin is removed and the back gear is meshed to spindle gear.

7. Thread cutting

- a) Reduce the speed spindle while thread cutting by using gear mechanism.

The bull gear pin is removed and the back gear is meshed to spindle gear.

- b) Select the combination of gears.

Use formula

Number of teeth in driver gear Number of teeth in driven gear

=Pitch to be cut/Lead screw pitch

= $5p_n/127$

Where p pitch to be cut

n- lead screw pitch

- c) Insert the selected driver and driven gear to tumbler gear mechanism.

- d) For engaging to cut left hand thread, tumbler gear lever must be pushed downwards. For right hand thread, the tumbler gear mechanism must operate towards the top.
- e) Engage the lead screw lever. In addition, engage lead screw to carriage by engaging split nut.
- f) Thread cutting tool and lock it into its position. Mount workpiece, cross slide such that the tip of the tool touches periphery of the workpiece. Make a mark on the graduation on the cross-slide hand wheel
- g) Rotate the cross-slide hand wheel by one division, which is equal to 0.2mm, thus, workpiece takes a depth of cut equal to 0.2mm. Now switch on the motor, which moves the carriage and simultaneously rotating workpiece.
- h) After the tool has reached the other end, the tool is brought back to previous position. Again, the cross-feed hand wheel is given a rotation by 2 divisions. Thus getting a depth of cut of 0.2mm. this sequence of operation is continued until correct thread form is cut on the profile.

Tabular column:

Sequence of Operations	Tools Used	Cutting Conditions			Measuring Instruments Used
		Speed	Depth of Cut	Feed	

Result:

Sl. No.	Dimensions		Deviation
	As per Drawing	Actual	

CALIBRATION OF MICROMETER

Aim: - To calibrate the micrometer using slip gauges to find periodic and progressive errors.

Apparatus used: Slip gauge set, micrometer.

Formulae: (i) Least Count = Pitch / no. of divisions on the head scale

(ii) Total Reading = PSR + (HSR x LC)

(iii) Error = $R_a - R_s$

(iv) % Error = $\frac{(R_a - R_s) \times 100}{R_s}$

Theory:

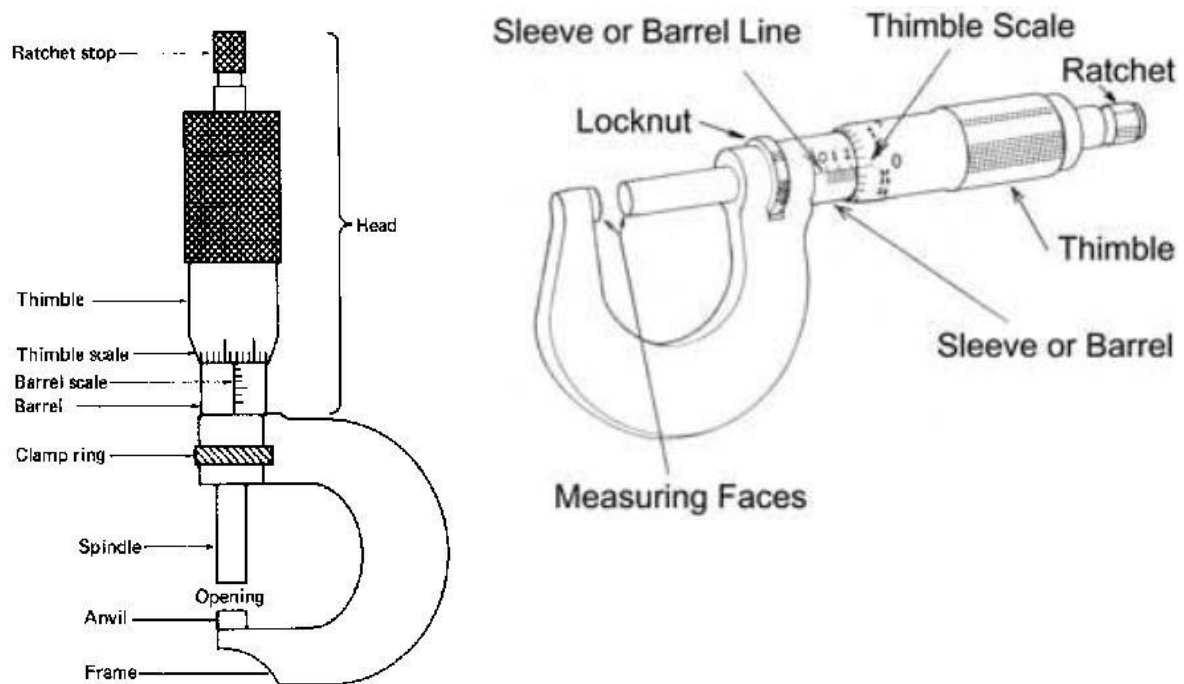


Figure 1.1 Outside Micrometer

A micrometer is a precision measuring instrument, used to obtain very fine measurements and available in metric and imperial versions. Line diagram of the Outside micrometer is shown in figure 1.1

Micrometers have two scales: a primary scale, on the barrel or sleeve, and a secondary scale, on the thimble. It is shown in figure 1.2.

Values are taken from each of these scales and combined to make the total measurement



Figure 1.2: Scales in Micrometer

There are different types of micrometer. The most common types are outside micrometers, inside micrometers and depth micrometers. The outside micrometer is the most commonly used type of micrometer. It is used to measure external dimensions such as the outside diameter of an object. Inside micrometers are used to measure inside dimensions like the inside diameter of a hole or tube. There are two types of inside micrometer: caliper-type inside micrometers and tubular and rod inside micrometers.

Slip Gauges

Gauge blocks (also known as gage blocks, Johansson gauges, slip gauges) are a system for producing precision lengths. Gauge blocks come in sets of blocks with a range of standard lengths.

They are used as a reference for the calibration of measuring equipment used in machine shops, such as micrometers, sine bars, calipers, and dial indicators (when used in an inspection role). Gauge blocks are the main means of length standardization used by industry.

Surface tension from oil that remains on the gage blocks or water vapor from the air that acts as an adhesive to hold them together. Molecular force caused by the interchange of electrons between the atoms of the separate blocks when two very flat surfaces are brought into such close contact with each other.

Least Count

The least count error is the error associated with the resolution of the instrument..

Least Count = Pitch of the micrometer/Total number of circular scale division

Error

The measurement error is defined as the difference between the true or actual value and the measured value. The true value is the average of the infinite number of measurements, and the measured value is the precise value.

Many types of errors are indicated while manufacturing micrometers. Pitch errors are most common among all errors. Pitch errors are errors in effective diameter of helix measured parallel to the axis of screw thread and are of mainly two types –progressive and periodic errors.

- 1) Progressive errors: If the pitch of the thread is uniform but is longer the shorter than its nominal value, such a type of errors is called as progressive errors. Progressive errors progressive in nature and increases as the length of axis increases. These errors occur when,
 - i) Tool work velocity ratio is incorrect.
 - ii) Change in length due to hardening by error in the pitch of lead screw.
 - iii) Due to fault in saddle guide ways.
 - iv) Due to casing of an incorrect gear train between work and tool.
- 2) Periodic errors: Periodic errors are those which vary in magnitude along the length of the thread and repeats at regular intervals. The deviation from the true helix occurs an adjacent threads at the same fraction of revolution but the pitch remains constant and such errors will not be detected by measurement along e line parallel to the axis. Errors of this type are most frequently caused by lack of squareness in the thrust bearing of the lead screw used to produce the thread. Some possible sources of errors which may result the incorrect functioning of the instrument are

- (i) Lack of flatness of the anvils.
- (ii) Lack of parallelism of the anvils.
- (iii) Inaccurate setting of the zero reading.
- (iv) Inaccurate reading followed by the zero position.
- (v) Inaccurate reading shown by the fractional divisions on the thimble.
- (vi) Errors in micrometer screw threads

Errors are also classified as systematic errors, random errors and blunders. Gross errors are caused by mistake in using instruments or meters, calculating measurement and recording data results.

Observations:

Type of Micrometer = _____
Make (Brand) = _____
Range = _____
Least count = _____

Type of Slip gauge set used = _____

Procedure:

1. Determine the least count of the micrometer by examining the micrometer.
2. Determine the zero error of the micrometer if any.
3. Place the required slip gauge between the jaws of the micrometer.
4. Clamp the gauge between the jaws of the micrometer.
5. Record the main scale reading.
6. Record the thimble scale reading and also the coinciding vernier.
7. Tabulate the results and repeat the procedure for the different slip gauges of various thicknesses.
8. Tabulate the results.
9. Calculate the actual reading and then calculate the progressive and periodic error for each of the trials and tabulate.
10. Plot the following graphs.
 - a. Actual reading v/s Standard reading
 - b. Periodic error v/s Standard reading
 - c. Progressive error v/s Standard reading

TABULAR COLUMN:

Sl no.	Zero error (ZE) (mm)	Slip gauge thickness (Rs) (mm)	Micrometer reading with slip gauge (Rm) (mm)	Actual reading (Ra) in mm $= Rm - ZE$	Periodic error $Pe = (Rs - Ra)/Rs \times 100$ (%)	Progressive error = $Pe_i - Pe_{(i+1)}$ (%)
1						-----
2						$Pe_1 - Pe_2$
3						
4						
5						
6						

Graphs

- Progressive error vs. Standard reading (Rs)
- Periodic error vs. Standard reading (Rs)
- Actual reading (Ra) vs. Standard reading (Rs)

Result: - The periodic and progressive errors for the various gauge thickness are determined and the necessary graphs are plotted.

CALIBRATION OF VERNIER CALIPER

Aim: To calibrate the given vernier calipers with respect to a standard reference i.e. slip gauge set.

Apparatus used: Slip gauge set, Venier Caliper

Theory: Calibrating a vernier caliper using slip gauges (gauge blocks) ensures measurement accuracy by comparing the tool against known, traceable standards. It involves cleaning the tools, checking for zero error, measuring several gauge blocks across the full range, calculating errors (Difference = Measured Value - True Size), and ensuring results are within tolerance.

Preparation

1. Clean and Inspect: Clean both the vernier caliper jaws and the slip gauges with cleaning solvent (like acetone) and a lint-free cloth.
2. Inspect for Damage: Check for smooth movement of the jaw and inspect for burrs or damage to the jaws.
3. Zero Error Check: Close the jaws gently and check if the zero on the vernier scale aligns with the zero on the main scale. Note any deviation as the "zero error".
4. Temperature Management: Ideally, perform the calibration in a controlled environment

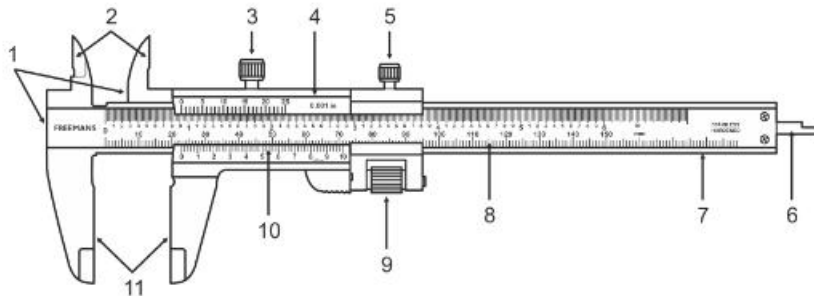
Procedure

1. Select Slip Gauges: Choose several standard slip gauges (e.g., 0.05 mm to 1.00 mm, and larger blocks like 50 mm, 100 mm) to cover the measuring range.
2. Measure and Record:
 - Wring together the slip gauges to make up the desired size, if necessary.
 - Place the slip gauges between the external measuring jaws.
 - Tighten the locking screw to secure the measurement.
 - Note the Main Scale Reading (MSR) and the Vernier Scale Reading (VSR).
3. Calculate Total Reading:
$$\text{Total reading} = \text{MSR} + (\text{VSR} \times \text{Least count})$$
4. Calculate Error:
$$\text{Error} = \text{Measured value} - \text{Slip guage size}$$
5. Repeat: Repeat the measurements at different positions of the jaws (e.g., near the tips and close to the beam) and at various points along the range.
6. Plot the following graphs.

- Actual reading v/s Standard reading
- Periodic error v/s Standard reading
- Progressive error v/s Standard reading

Key Precautions

- Wringing: Ensure slip gauges are properly wrung together to avoid errors due to gaps.
- Force: Apply consistent and light pressure on the jaws to avoid bending the caliper jaw.
- Temperature: Allow slip gauges to reach the room temperature of the laboratory before testing to prevent thermal expansion errors.



- | | |
|--------------------------------------|---------------------------------------|
| ① Step Measuring Faces | ⑦ Beam |
| ② Inside Measuring Faces | ⑧ Main Scale |
| ③ Clamping Screw | ⑨ Adjusting Screw for Fine Adjustment |
| ④ Slider | ⑩ Vernier Scale |
| ⑤ Clamping Screw for Fine Adjustment | ⑪ Outside Measuring Faces |
| ⑥ Depth Measuring Blade | |

Least Count

Least Count = value of smallest main scale division / Total number of divisions on vernier scale

Observations:

Type of Vernier = _____
Make (Brand) = _____
Range = _____
Least count = _____

Type of Slip gauge set used = _____

TABULAR COLUMN:

Sl no.	Zero error (ZE) (mm)	Slip gauge thickness (Rs) (mm)	Vernier reading with slip gauge (Rm) (mm)	Actual reading (Ra) in mm =Rm - ZE	Periodic error Pe = (Rs-Ra)/Rs*100 (%)	Progressive error = Pe _i - Pe _(i+1) (%)
1						-----
2						Pe ₁ - Pe ₂
3						
4						
5						
6						

Graphs

- Progressive error vs. Standard reading (Rs)
- Periodic error vs. Standard reading (Rs)
- Actual reading (Ra) vs. Standard reading (Rs)

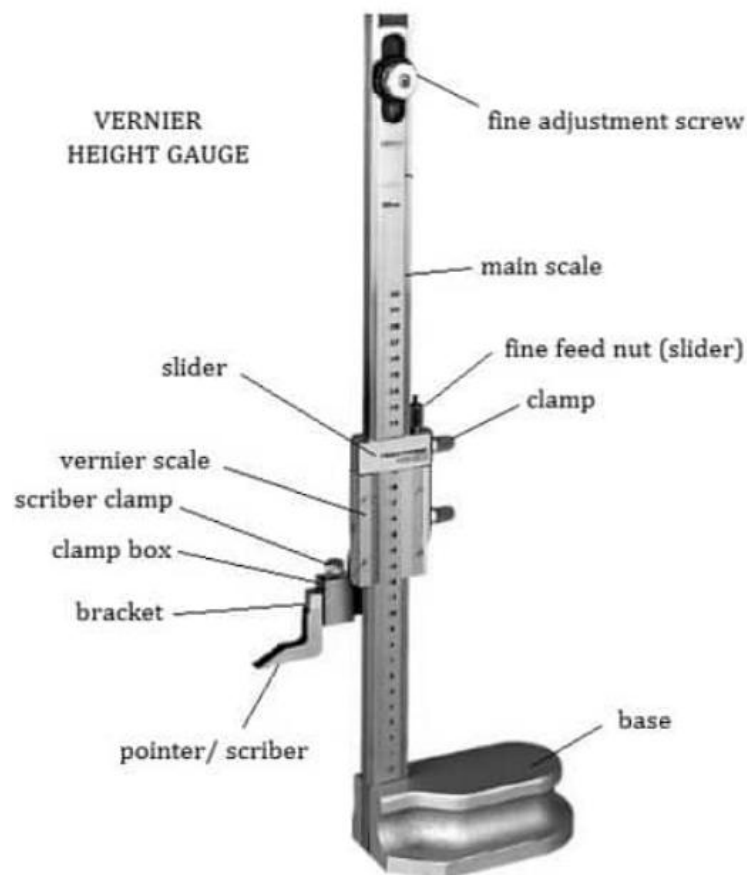
Result: - The periodic and progressive errors for the various gauge thickness are determined and the necessary graphs are plotted.

CALIBRATION OF VERNIER HEIGHT GAUGE

AIM: To calibrate the given vernier height gauge with respect to a standard reference i.e. slip gauge set.

Apparatus used: Slip gauge set, Venier Height Gauge

Theory: Calibrating a Vernier height gauge using slip gauges involves measuring precision-ground blocks on a surface plate, then comparing the gauge reading to the actual block size to find errors. First, clean and set the height gauge to zero on the surface plate. Then, measure several slip gauge combinations (e.g., 10, 25, 50, 75mm) within its range to detect measurement errors across its scale.



Procedure

1. Preparation: Clean the surface plate, vernier height gauge, and slip gauge set to remove oil and dust.
2. Zero Setup: Place the height gauge on the surface plate and lower the scriber to touch the plate. Set the main scale and vernier to zero (or note zero error).
3. Arrange Slip Gauges: Build up various sizes using slip gauges (e.g., 10 mm, 50 mm, 100 mm) through a process called ringing.

4. Measurement: Place the slip gauge stack on the surface plate beneath the scriber. Move the gauge down until the scriber touches the top of the slip gauge.
5. Reading: Take the reading, ensuring there is no parallax error (use a magnifying glass if provided).
6. Calculate Total Reading:
$$\text{Total reading} = \text{MSR} + (\text{VSR} \times \text{Least count})$$
7. Calculate Error:
$$\text{Error} = \text{Measured value} - \text{Slip gauge size}$$
8. Repeated Testing: Repeat the process for at least 5-10 different measurements covering the entire range of the height gauge
7. Plot the following graphs.
 - Actual reading v/s Standard reading
 - Periodic error v/s Standard reading
 - Progressive error v/s Standard reading

Key Precautions

- Wringing: Ensure slip gauges are properly wrung together to avoid errors due to gaps.
- Force: Apply consistent and light pressure on the jaws to avoid bending the caliper jaw.
- Temperature: Allow slip gauges to reach the room temperature of the laboratory before testing to prevent thermal expansion errors.

Least Count

Least Count = value of smallest main scale division / Total number of divisions on vernier scale

Observations:

Type of Vernier = _____
Make (Brand) = _____
Range = _____
Least count = _____

Type of Slip gauge set used = _____

TABULAR COLUMN:

Sl no.	Zero error (ZE) (mm)	Slip gauge thickness (Rs) (mm)	Vernier reading with slip gauge (Rm) (mm)	Actual reading (Ra) in mm =Rm - ZE	Periodic error Pe = (Rs-Ra)/Rs*100 (%)	Progressive error = Pe _i - Pe _(i+1) (%)
1						-----
2						Pe ₁ - Pe ₂
3						
4						
5						
6						

Graphs

- Progressive error vs. Standard reading (Rs)
- Periodic error vs. Standard reading (Rs)
- Actual reading (Ra) vs. Standard reading (Rs)

Result: - The periodic and progressive errors for the various gauge thickness are determined and the necessary graphs are plotted.

CLEAVELAND (OPEN CUP) FLASH & FIRE POINT APPARATUS

Aim:

To determine the flash and fire points of a given oil sample.

Apparatus Required:

1. Cleaveland Open Cup apparatus,
2. Sample Oil,
3. Thermometer.

Procedure:

1. Oil cup is cleaned and then given oil sample is poured into it up to the filling mark,
2. The oil under test is heated and heating process is accompanied with continuous stirring which ensures uniform distribution of heat,
3. The test flame is introduced for 5°C rise in temperature. No stirring is done during the checking of test flame,
4. If a distinct flash is visible in the two observation parts, record the temperature. This corresponds to flash point,
5. This is an open type heating and hence for every 2°C temperature rise, test flame is introduced and point at which all oil ignites is called fire point.
6. Heating is stopped and the apparatus is allowed to cool.

Sketch:

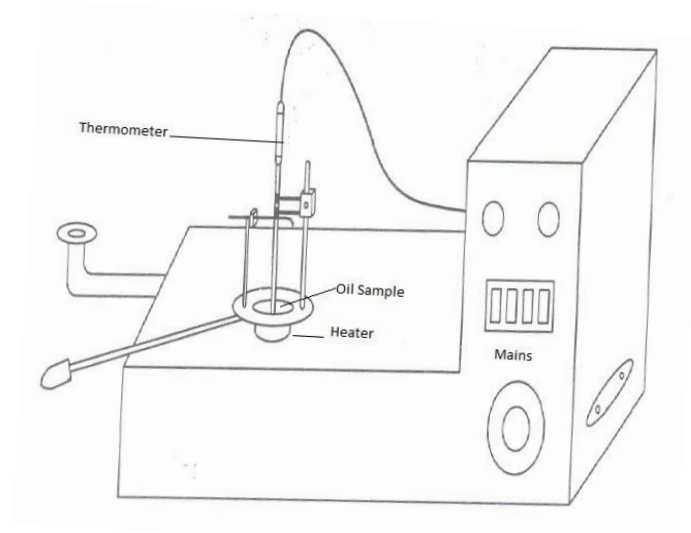


Figure-2: Cleaveland open cup apparatus

Observation Table:

S.NO	Temperature of Oil	Inference
Units	(°C)	
1.		
2.		
3.		
4.		
5.		

Result:

Flash Point of given oil Sample is _____°C,
Fire Point of given oil Sample is _____°C.

Applications:

Remarks:

REDWOOD VISCOMETER

Aim:

To determine the viscosity of the given specimen oil (Diesel Oil) at different temperatures.

Apparatus Required:

1. Redwood Viscometer,
2. Thermometer,
3. Collecting Flask (50 ml),
4. Stop Clock,
5. Diesel oil.

PROCEDURE:

1. The weight balance is set to zero reading,
2. The specimen oil is poured into the flask and its initial temperature is found out, also the mass of the flask with the oil is noted down,
3. Time taken by the specimen oil to fill 50ml flask is noted down. During the flow the stop wire is lifted up,
4. Set the voltage to 150V and start reading the heated oil. Note down the time taken to collect 50ml of oil for every 10°C rise in temperature. Also weigh the flask with oil,
5. Then determine kinematic, dynamic viscosity and redwood number.

Sketch:

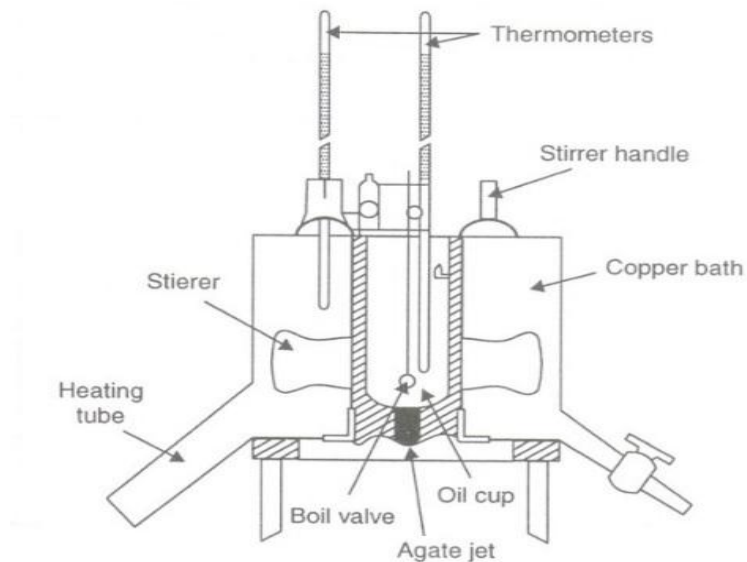


Figure-3: Redwood viscometer apparatus

Formulae Used:

1. Kinematic Viscosity

$$\gamma = \left[0.246 T - \left(\frac{190}{T} \right) \right] \times 10^{-6} \text{ m}^2/\text{Sec}$$

2. Redwood Number

$$\text{RWN} = \left[\frac{100 \times T \times S}{535 \times 0.915} \right]$$

3. Specific gravity,

$$\text{Specific Gravity} = \left[\frac{W_2 - W_1}{50} \right]$$

Where, W_1 = Weight of empty flask, W_2 = Weight of flask with oil.

4. Density of Fuel = Specific gravity x Density of Water,

5. Dynamic viscosity, $\mu = \rho_{\text{Fuel}} \times \gamma_{\text{Fuel}}$

TABULAR COLUMN:

S.NO	Temp of Oil	Time for Collecting oil 'T'	Weight of Oil in Flask 'W ₂ '	Specific Gravity 'S'	Redwood Number	Kinematic Viscosity	Dynamic Viscosity
Units	°C	Sec	Gm			m^2/s	Ns/m^2
1							
2							
3							
4							

GRAPHS:

1. Kinematic Viscosity vs Temperature,
2. Dynamic viscosity vs Temperature.

RESULT: The Kinematic Viscosity and Dynamic Viscosity of the given Specimen oil are determined.

Applications:**Remarks:**

S.I. SYSTEM**Fundamental units of S.I system**

Sr. No.	Physical quantities	Unit	symbol
1	Length	Metre	m
2	Mass	Kilogram	Kg
3	Time	Second	S
4	Temperature	Kelvin	K

Supplementary units of S.I. system

Sr. No.	Physical quantities	Unit	symbol
1	Plane angle	Radian	Rad

Principal S.I. units

Sr. No.	Physical quantities	Unit	symbol
1	Force	Newton	N
2	Work	Joule	J, N.m
3	Power	Watt	W
4	Energy	Joule	J, N.m
5	Area	Square metre	m ²
6	Volume	Cubic metre	m ³
7	Pressure	Pascal	Pa
8	Velocity/speed	metre per second	m/s
9	Acceleration	metre/second ²	m/s ²
10	Angular velocity	radian/second	rad/s
11	Angular acceleration	radian/second ²	rad/s ²
12	Momentum	kilogram metre/second	Kg.m/s
13	Torque	Newton metre	N.m
14	Density	Kilogram/metre ³	Kg/m ³
15	Couple	Newton.metre	N.m
16	Moment	Newton.metre	N.m

S.I. Prefixes

Multiplication factor	Prefix	Symble
10 ¹²	Tera	T
10 ⁹	Giga	G
10 ⁶	Mega	M
10 ³	kilo	k
10 ²	hecto	h
10 ¹	deca	da
10 ⁻¹	deci	d
10 ⁻²	centi	c
10 ⁻³	milli	m
10 ⁻⁶	micro	μ
10 ⁻⁹	nano	n
10 ⁻¹²	pico	p

UNIT CONVERSION	
1 m = 100 cm = 1000 mm	1 Mpa = 1 N/mm ²
1 km = 1000 m	1 Gpa = 10 ³ N/mm ²
1 cm ² = 100 mm ²	1 Pascal = 1 N/m ²
1 m ² = 10 ⁶ mm ²	1 degree = $\frac{\pi}{180}$ radians
1 kgf = 9.81 N = 10 N	
1 kN = 10 ³ N	

QUANTITY					
Scalar Quantity: “A Scalar Quantity is one which can be completely specified by its magnitude only”			Vector Quantity: “A vector Quantity is one which requires magnitude and direction both to completely specified it”		
Length	Mass	Distance	Displacement	Force	Weight
Density	Area	Temperature	Velocity	Angular displacement	Acceleration
Volume	Speed	Time	Angular velocity	Momentum	Angular acceleration
Energy	Work	Moment of inertia	Moment	impulse	

Space: It is a region in all directions encompassing the universe. It is a geometric position occupied by bodies. These positions are describe by linear or angular measurements with reference to a defined system of co-ordinates.

Time: Time is a measurement to measure a duration between successive events. In the study of statics time does not play important role. In dynamics time is very important parameter. In all system of units, unit of time is second.

Particle: A particle is ideally dimensionless. But it has a very small mass.

Rigid body: No body is perfectly rigid, however rigid body is defined as a body in which particles do not change their relative positions under the action of any force or torque. Rigid body is ideal body. When the body does not deform under the action of A force or A torque, body is said rigid.

Deformable body: When a body deforms due to A force or A torque it is said deformable body. Material generates stresses against deformation.

Force: Force is an agent, which generates or tends to generate and destroy or tends to destroy the motion in a body.

Characteristics of a force:

- It has a magnitude
- It has a direction
- It is a vector quantity
- It has a point of application
- It has a nature
 - Tensile force
 - Compressive force
 - Pull force
 - Push force

SYSTEM OF FORCES

When two or more forces act on a body, they are called to for a system of forces.

Coplanar forces: The forces, whose lines of action lie on the same plane, are known as coplanar forces.

Collinear forces: The forces, whose lines of action lie on the same line, are known as collinear forces.

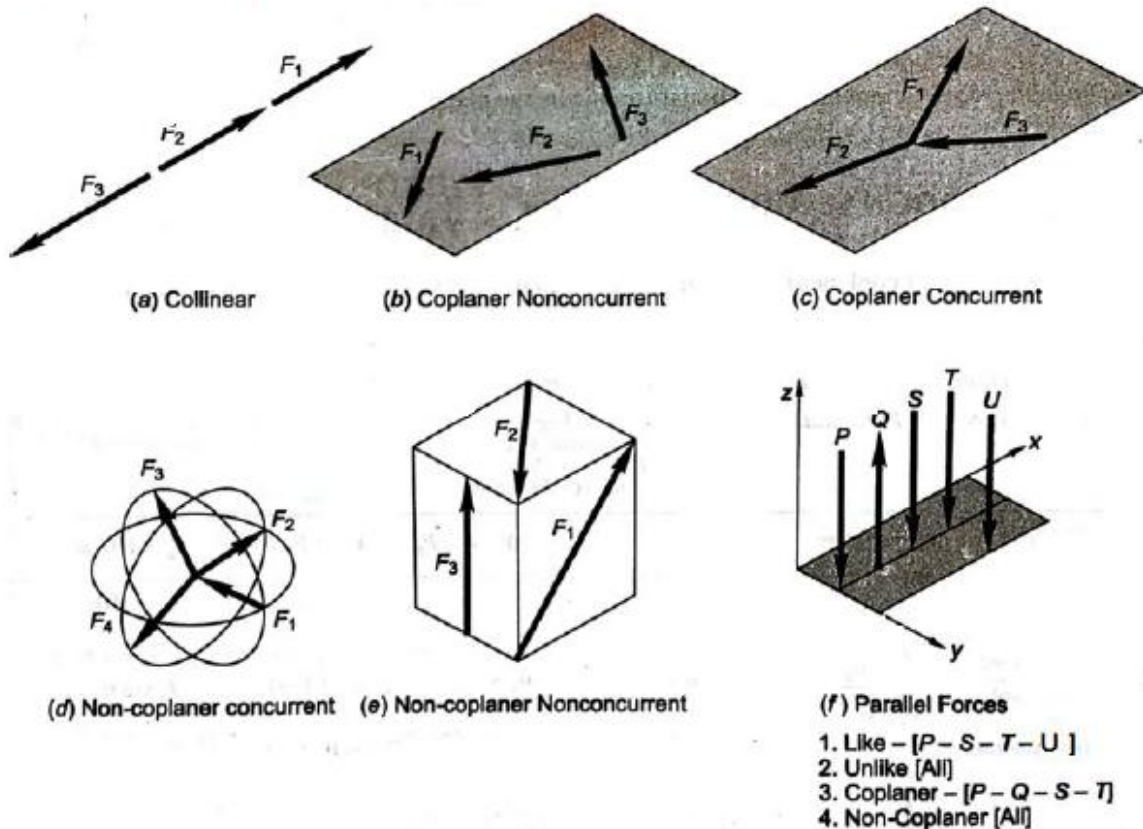
Concurrent forces: The forces, which meet at one point, are known as concurrent forces. The concurrent forces may or may not be collinear.

Coplanar concurrent forces: The forces, which meet at one point and their line of action also lay on the same plane, are known as coplanar concurrent forces.

Coplanar non-concurrent forces: The forces, which do not meet at one point, but their lines of action lie on the same, are known as coplanar non-concurrent forces.

Non-Coplanar concurrent forces: The forces, which meet at one point, but their lines of action do not lie on the same plane, are known as non-coplanar concurrent forces.

Non-Coplanar non-concurrent forces: The forces, which do not meet at one point and their lines of action do not lie on the same plane, are called non-coplanar non-concurrent forces.



➤ Force:

“An agent which produces or tends to produce, destroys or tends to destroy motion of body is called force”

Unit: N, kN, Kg etc.

Quantity : Vector

Characteristics of Force:

- 1) **Magnitude:** Magnitude of force indicates the amount of force (expressed as N or kN) that will be exerted on another body
- 2) **Direction:** The direction in which it acts
- 3) **Nature:** The nature of force may be tensile or compressive
- 4) **Point of Application:** The point at which the force acts on the body is called point of application

Types of Forces:

- Contact Force
- Body force
- Point force and distributed force
- External force and internal force
- Action and Reaction
- Friction force
- Wind force
- Hydrostatic force
- Cohesion and Adhesion
- Thermal force

System of Forces:

- Coplanar Forces
- Concurrent forces
- Collinear forces
- Coplanar concurrent forces
- Coplanar non-concurrent forces
- Non-coplanar concurrent forces
- Non-coplanar non-concurrent forces
- Like parallel forces
- Unlike parallel forces
- Spatial forces

➤ Principle of Individual Forces

1) Principle of transmissibility:

“If a force acts at a point on a rigid body, it may also be considered to act at any other point on its line of action, provided the point is rigidly connected with the body.”

2) Principle of Superposition of forces:

“If two equal, opposite and collinear forces are added to or removed from the system of forces, there will be no change in the position of the body. This is known as principle of superposition of forces.”

COPLANAR CONCURRENT FORCES

Resultant Force:

If number of Forces acting simultaneously on a particle, it is possible to find out a single force which could replace them or produce the same effect as of all the given forces is called resultant force.

Methods of Finding Resultant:-

- 1) Parallelogram Law of Forces (For 2 Forces)
- 2) Triangle Law (For 2 Forces)
- 3) Lami's theorem (For 3 forces)
- 4) Method of resolution (For more than 2 Forces)

[1]	Parallelogram law of forces $R = \sqrt{P^2 + Q^2 + 2PQ \cos \theta}$ $\tan \alpha = \frac{Q \sin \theta}{P + Q \cos \theta}$ Where, R = Resultant force θ = angle between P and Q α = angle between P and R	
[2]	Triangle law of forces $R = \sqrt{P^2 + Q^2 - 2PQ \cos \beta}$ Where, $180^\circ - \theta$ R = Resultant force θ = angle between P and Q α = angle between P and R $\alpha = \sin^{-1} \left(\frac{Q}{R} \sin \beta \right)$	
[3]	Lami's theorem $\frac{P}{\sin \alpha} = \frac{Q}{\sin \beta} = \frac{R}{\sin \gamma}$ Where, P, Q, R are given forces α = angle between Q and R β = angle between P and R γ = angle between P and Q	
[4]	Resolution of concurrent forces $\sum H = P_1 \cos \theta_1 + P_2 \cos \theta_2 + P_3 \cos \theta_3 + P_4 \cos \theta_4$ $\sum V = P_1 \sin \theta_1 + P_2 \sin \theta_2 + P_3 \sin \theta_3 + P_4 \sin \theta_4$ $R = \sqrt{(\sum H)^2 + (\sum V)^2}$ $\tan \theta = \frac{\sum V}{\sum H}$ Where, P_1, P_2, P_3, P_4 are given forces $\theta_1, \theta_2, \theta_3, \theta_4$ are angle of accordingly P_1, P_2, P_3, P_4 forces from X-axes R = Resultant of all forces θ = angle of resultant with horizontal	

Applied Mechanics

Forces, Vectors and Resultants

As a scientific discipline, applied mechanics derives many of its principles and methods from the physical sciences (in particular, mechanics and classical mechanics), from mathematics and, increasingly, from computer science. As such, applied mechanics shares similar methods, theories, and topics with applied physics, applied mathematics, and computer science.

Applied mechanics, as its name suggests, bridges the gap between physical theory and its application to technology. As such, applied mechanics is used in many fields of engineering, especially mechanical engineering. In this context, it is commonly referred to as engineering mechanics. Much of modern engineering mechanics is based on Isaac Newton's laws of motion while the modern practice of their application can be traced back to Stephen Timoshenko, who is said to be the father of modern engineering mechanics.

Applied mechanics in engineering

Typically, engineering mechanics is used to analyze and predict the acceleration and deformation (both elastic and plastic) of objects under known forces (also called loads) or stresses.

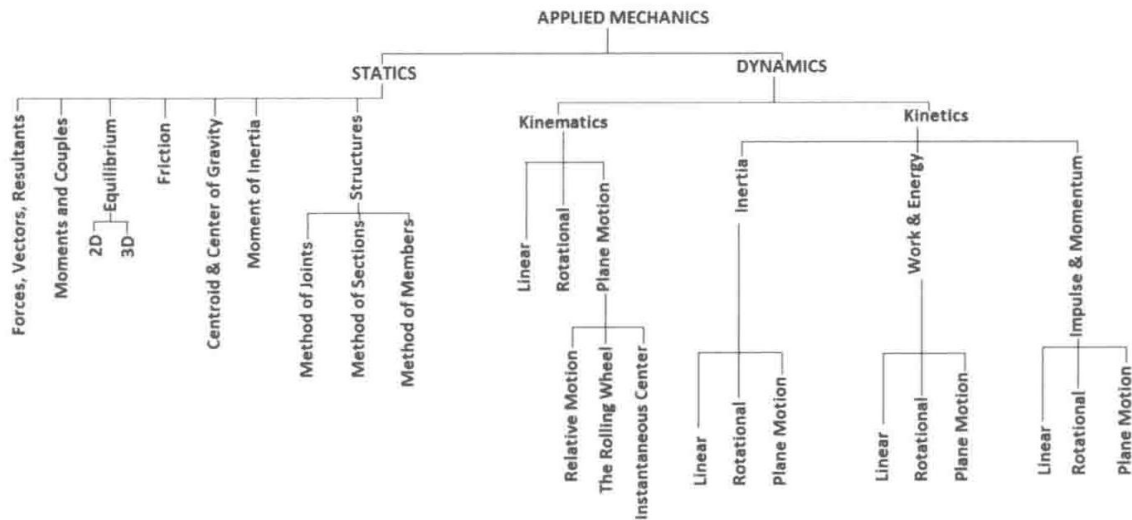
When treated as an area of study within a larger engineering curriculum, engineering mechanics can be subdivided into:

- Statics, the study of non-moving bodies under known loads.
- Dynamics, the study of how forces affect moving bodies.
- Mechanics of materials or strength of materials, the study of how different materials deform under various types of stress.
- Deformation mechanics, the study of deformations typically in the elastic range.
- Fluid mechanics, the study of how fluids react to forces. Note that fluid mechanics can be further split into fluid statics and fluid dynamics, and is itself a sub-discipline of continuum mechanics. The application of fluid mechanics in engineering is called hydraulics.
- Continuum mechanics is a method of applying mechanics that assumes that all objects are continuous. It is contrasted by discrete mechanics and the finite element method.

Scope of Applied Mechanics

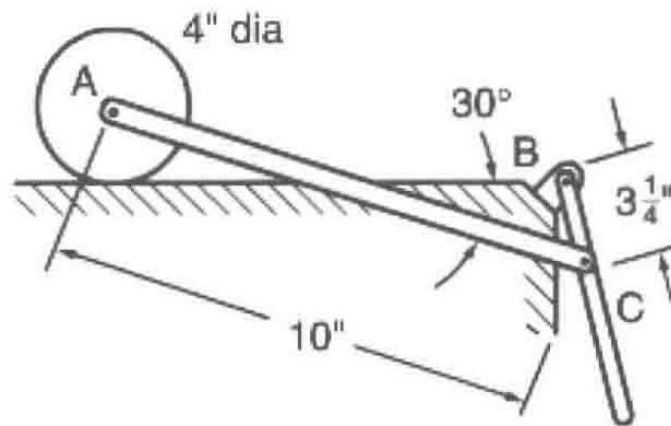
The scope of applied mechanics covered in this textbook is shown in Figure 1.1, which embraces virtually all topics covered in standard college and university curriculum in this subject. The new approach followed in this textbook is to apply SolidWorks to the entire spectrum of Figure 1.1.

Courtesy: Applied Mechanics using Solidworks – Godfrey Onwubolu Imperial College Press

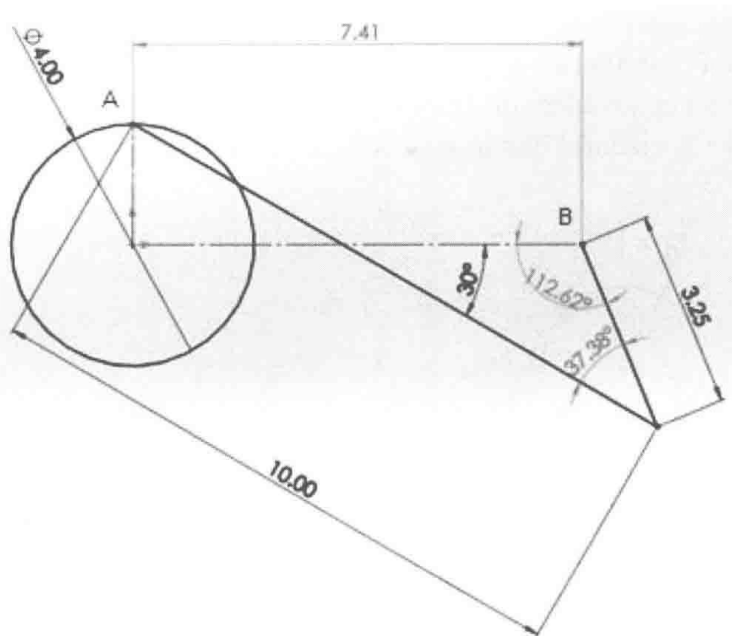


Problem 1

A roller and lever mechanism is in the position shown in Figure. Find the horizontal distance between A and B. (Point B is at the same level as the horizontal surface.)



Solution:



Sketch the given problem description (See Figure above).

Measure the horizontal distance of AB.

The horizontal distance between A and B = 7.41 in.

Vector Addition: Graphical

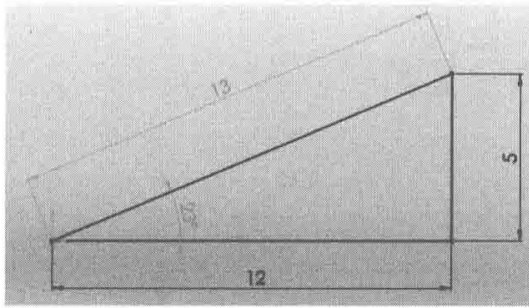
The sum of two or more vectors is a resultant. Graphical vector addition requires the drawing of the vector to some scale in their given direction. The resultant can then be measured or scaled from the drawing.

There are three known methods or rules of vector addition:

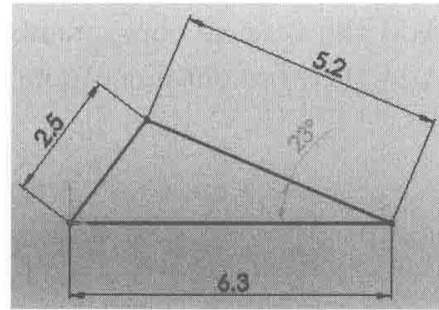
Triangle: Used when forces form a right-angle triangle or any other triangle (Figure (a) and (b))

Parallelogram: The diagonal of the parallelogram formed represent the resultant (Figure (c))

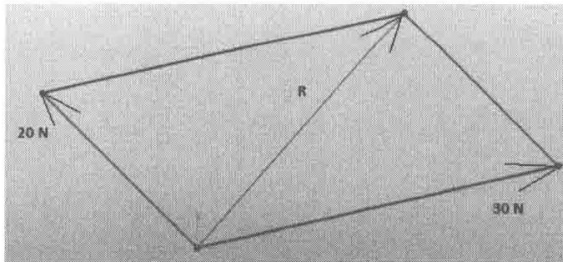
Vector polygon: A continuation of the triangle rule to accommodate more than two forces (Figure (d))



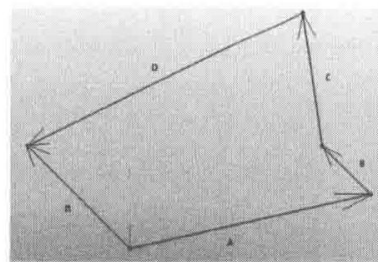
(a) Right-angle triangle.



(b) Other triangle.



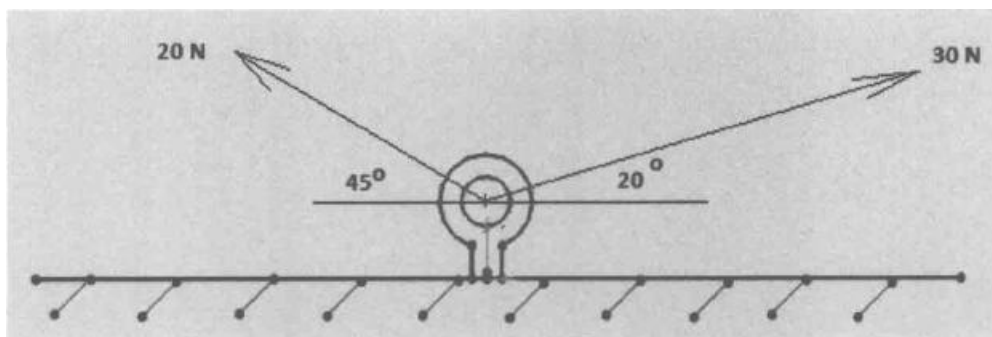
(c) Parallelogram.



(d) Vector polygon.

Problem 2

Forces of 20 N and 30 N are pulling on a ring as shown in the figure. Determine the resultant using the triangle rule.



Solution:

1. Open a New Part Document.
2. Select the Front Plane.
3. Sketch a line parallel to the 30 N force at an angle of 30° to the horizontal.
4. At the end of the first sketch, sketch another line parallel to the 20 N force at an angle of 45° to the horizontal.

5. Sketch a line from the end of the first line to the end of the second line (see Figure 2.6). (This last line is the resultant of the two forces with a value of 28 N.)

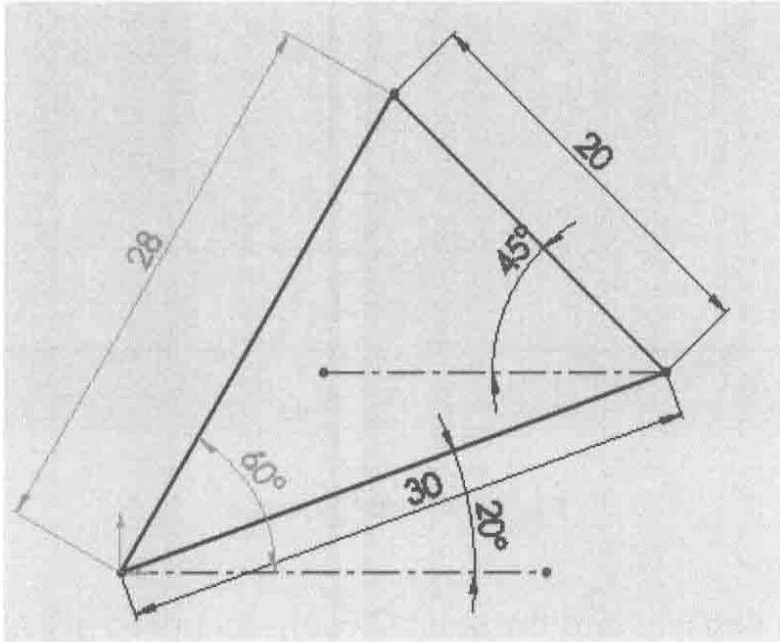
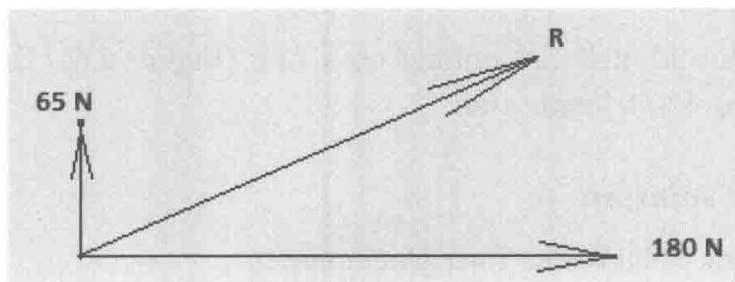


Fig. 2.6 Resultant of two forces.

Problem 3

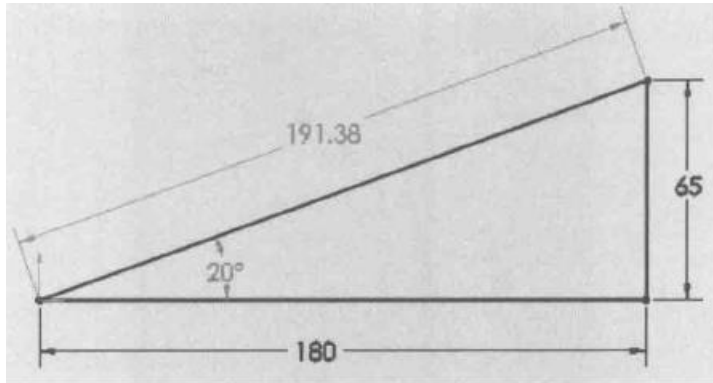
Determine the resultant of the vectors shown in Figure.



Solution:

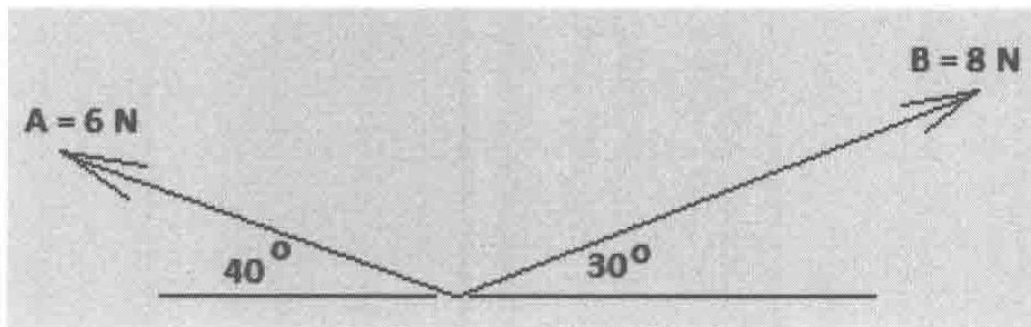
1. Open a New Part Document.
2. Select the Front Plane.
3. Sketch a line parallel to the 180 N force that is horizontal.

4. At the end of the first sketch, sketch another line parallel to the 65 N force at an angle of 90° to the horizontal.
5. Sketch a line from the end of the first line to the end of the second line (see Figure below). (This last line is the resultant of the two forces; with a value of 191.38 N. at an angle of 20° to the horizontal.)



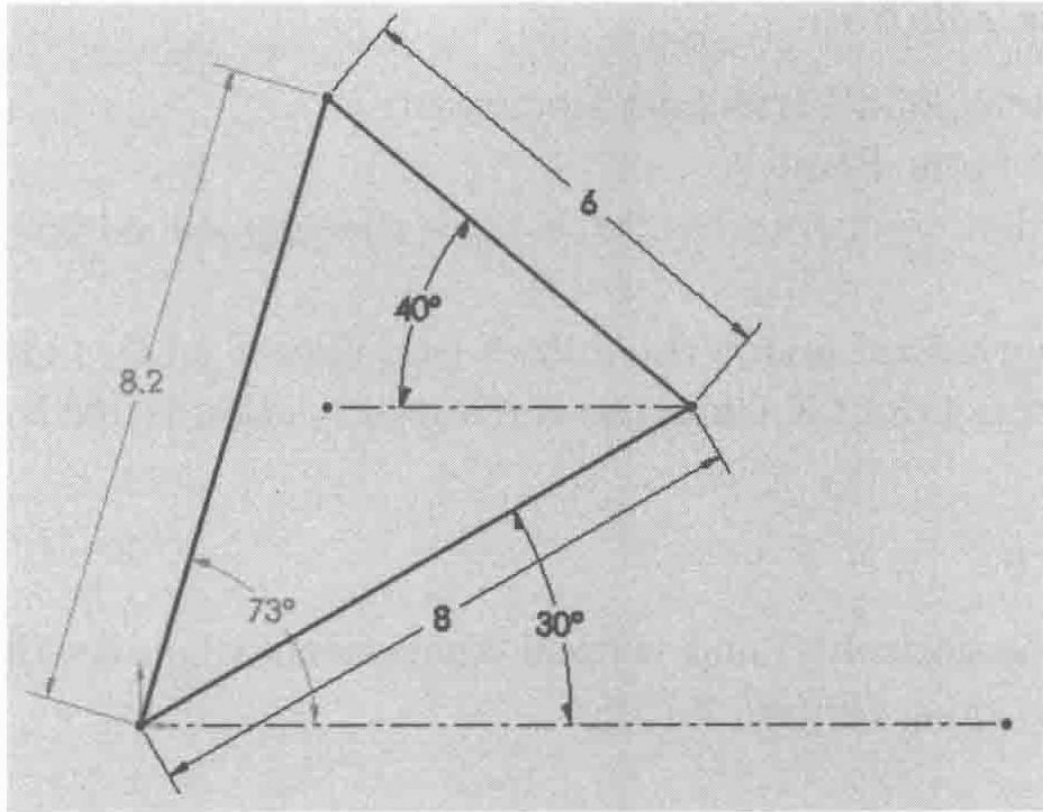
Problem 4

Find the resultant of the force system shown in figure



Solution

1. Open a New Part Document.
2. Select the Front Plane.
3. Sketch a line parallel to the 8 N force that is inclined 30° to the horizontal.
4. At the end of the first line, sketch another line parallel to the 6 N force at an angle of 40° to the horizontal.
5. Sketch a line from the end of the first line to the end of the second line (see Figure below). (This last line is the resultant of the two forces; with a value of 8.2 N, at an angle of 73° to the horizontal.)



Moments and Couples

Moment of a Force

Moment is the same as a couple, which is the measure of the turning effect, or torque, produced by a force, acting on a body. Examples include pushing a revolving door while walking through it, tightening of a nut with a wrench, turning a steering wheel of a car, unscrewing a fluorescent bulb from its fitting, etc. If a force is acting some distance away from a point, it causes a twisting action about the point. This twisting action, or torque, is called a moment. The magnitude of the moment depends upon both the size of the force and the perpendicular distance from the force to the point. Mathematically, we can define the moment as:

Moment = force $\times$ the perpendicular distance between the axis and the line of action of the force.

Moments are vector quantities; therefore, their direction must be specified. Figure shows the force F and the perpendicular distance d from O for determining moment/torque.

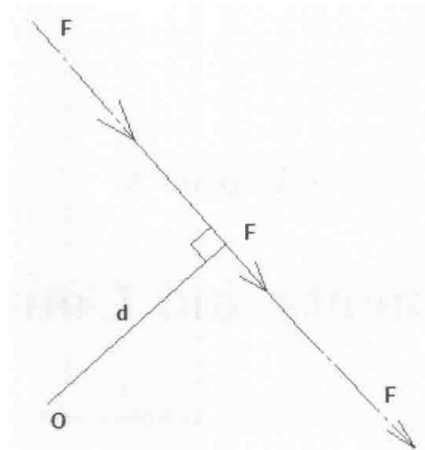


Fig: Force and perpendicular distance

Couples

A couple consists of two equal forces, acting in opposite directions and separated by a perpendicular distance. When you turn your steering wheel with both hands, equal forces are applied and these are separated by the diameter of the steering wheel. Figure shows the force F and the perpendicular distance d for determining the couple.

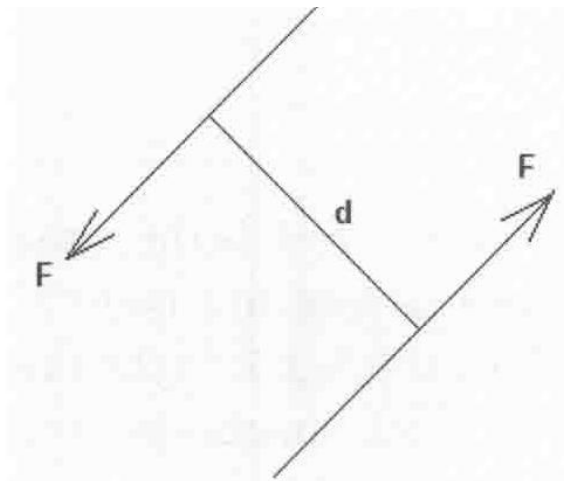


Fig: Couple - Force and perpendicular distance

a couple is a specific system of forces that produces rotation only, with no linear motion. Because the forces are equal and opposite, the net force is zero. This means the object will not move left, right, up, or down. However, because the forces are offset, they create a turning effect called a couple moment.

Couple Moment (M)

$$M = F \times D$$

F = The magnitude of one of the forces (since they are both the same).

D = The perpendicular distance (often called the arm of the couple) between the two lines of action.

Problem: Determine the moment about point A of the lever shown in Figure.

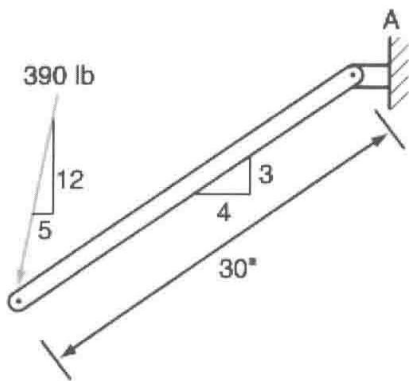
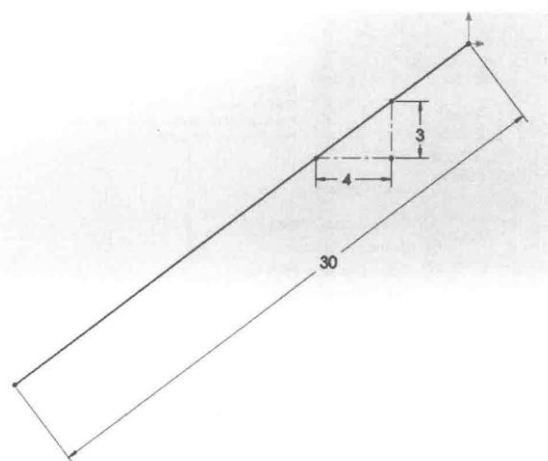


Fig: Lever and load at its end.

Select Front Plane.

Be in Sketch mode.

Create the Sketch as shown in Figure



Two-Dimensional Equilibrium

Free-Body Diagrams

FBDs are diagrams of objects in static (not moving) equilibrium (forces acting on the object are balanced against each other so that the object does not move). A FBD shows an object or body with all supports removed and replaced by forces in balance. All forces acting on an object must be shown.

One mistake that students make is to retain the fixtures supporting the body for which a FBD is sought. The guiding principle is that a free-body is indeed 'free'! Students normally enjoy this definition, but end up making mistake when they are drawing FBDs. FBDs must float otherwise they cannot be free-body diagrams.

A FBD of a member is a picture showing how the rest of the world is acting *on* the member, not what the member is doing *to* anything else. This concept is important; visualize yourself as taking the place of a member and consider how all other forces around you are acting on you.

Another pitfall with students is that they do not obey the equations set out for equilibrium. Let us consider the equilibrium equations:

$$\sum F_x = 0; \quad \sum F_y = 0; \quad \sum M = 0.$$

The forces must be summed up with appropriate signs and equated to 0; so also the moment about a joint of the mechanism must be summed up with appropriate signs and equated to 0. If this is not done correctly, it leads to an error. Therefore, it is recommended that the mathematical definitions for equilibrium be enforced when writing the equilibrium equations. Doing things correctly is a good practice that leads to correct solutions.

The other important point to note is that the FBD conventions are extremely useful.

Free-body diagram conventions

There are conventions for replacing supports with equivalent supporting forces as shown in Figure 4.1.

1. *Roller*: The roller cannot exert a horizontal force; therefore, only a force perpendicular to the surface is present (see 1 in Figure 4.1).
2. *Roller*: The only roller force is that perpendicular to the roller surface (see 2 in Figure 4.1).
3. *Smooth Surface*: Zero friction is assumed; therefore, only one force, that perpendicular to the surface, is present (see 3 in Figure 4.1).

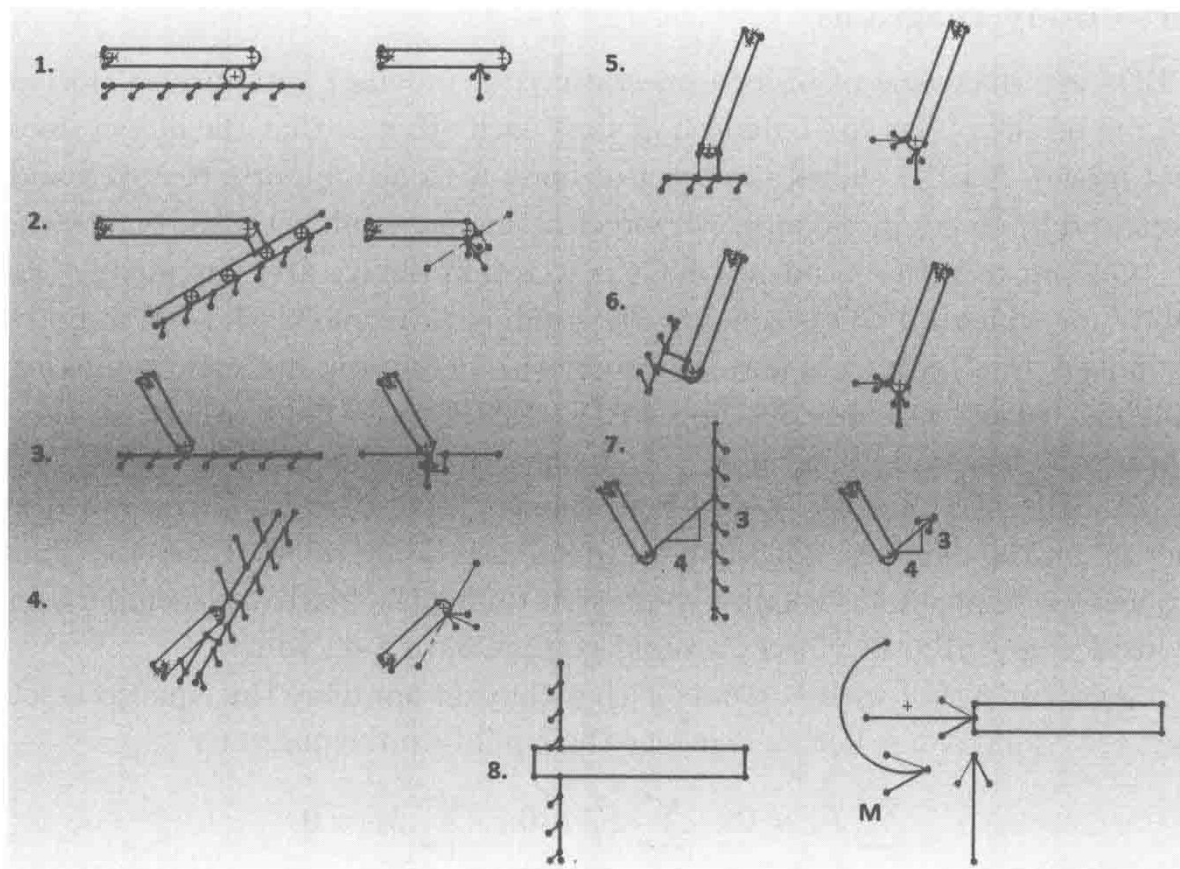


Fig: Conventions for replacing supports with equivalent supporting forces

4. *Slot*: The same principle applies as for a smooth surface. There is only one force present, that perpendicular to the slot (see 4 in Figure 4.1).
5. *Pinned*: Both horizontal and vertical components must be assumed at a pinned connection unless it is on a roller or smooth surface (see 5 in Figure 4.1).
6. The orientation of the support is immaterial; horizontal and vertical components must be assumed (see 6 in Figure 4.1).
7. *Cable*: This is always a single force pulling in the direction of the cable (see 7 in Figure 4.1).
8. *Fixed Support*: The beam is embedded in the wall or support and therefore has three possible reactions: A moment, vertical force, and horizontal force (see 8 in Figure 4.1).

Three Equations of Equilibrium

For complete static equilibrium, three requirements must be met:

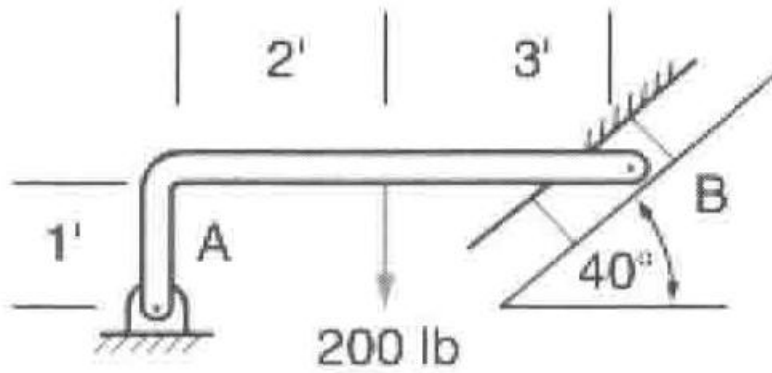
1. Horizontal forces balance.
2. Vertical forces balance.
3. Moments balance; clockwise = counter-clockwise (about any point).

Mathematically, these conditions can be stated as follows:

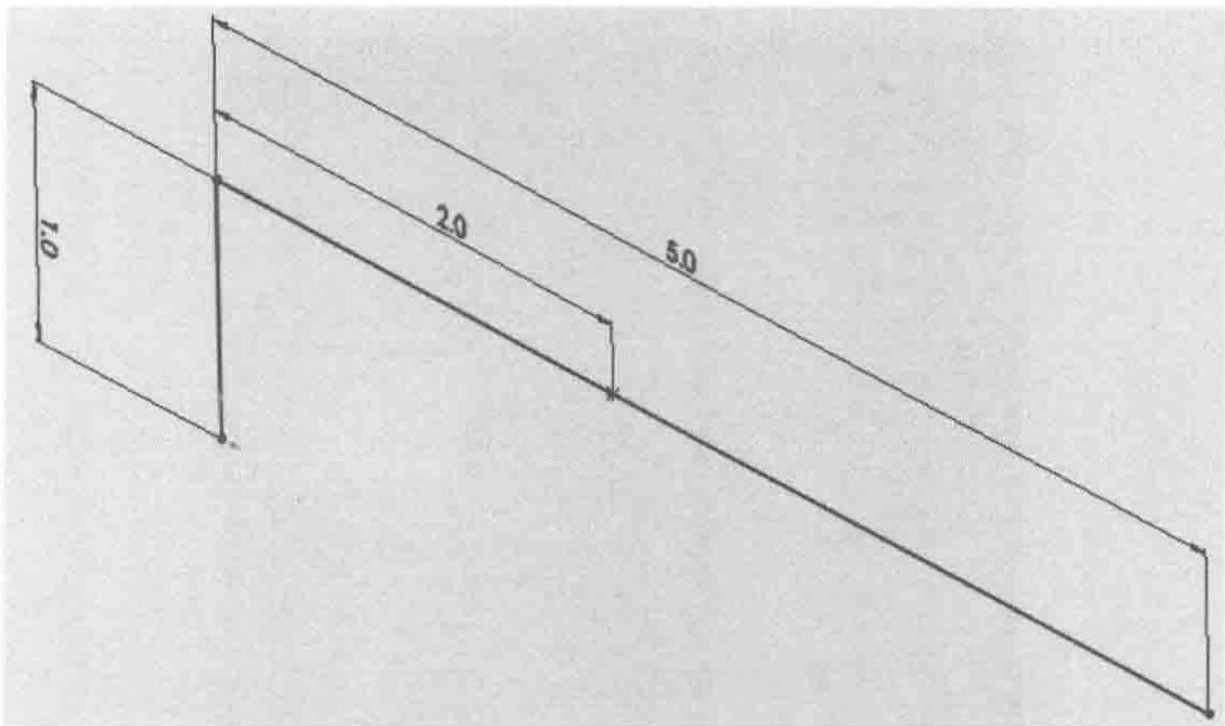
1. $\sum F_x = 0$.
2. $\sum F_y = 0$.
3. $\sum M = 0$.

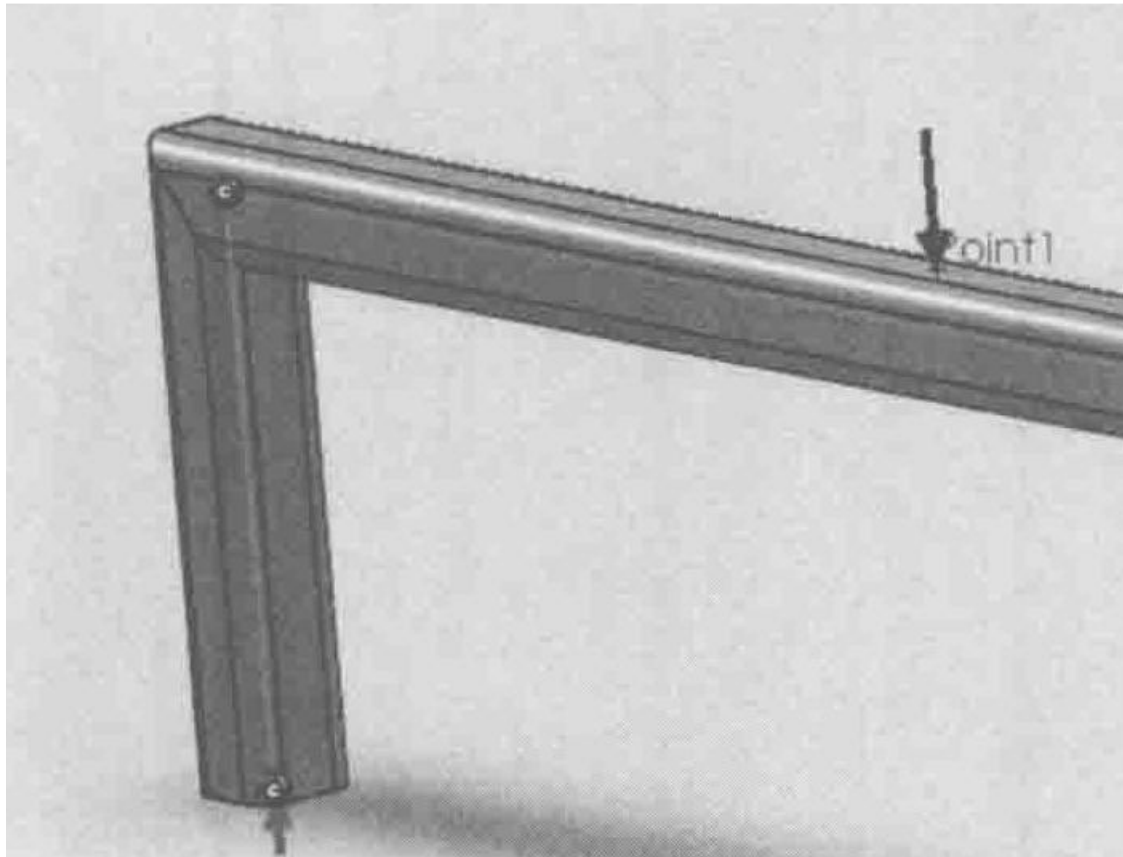
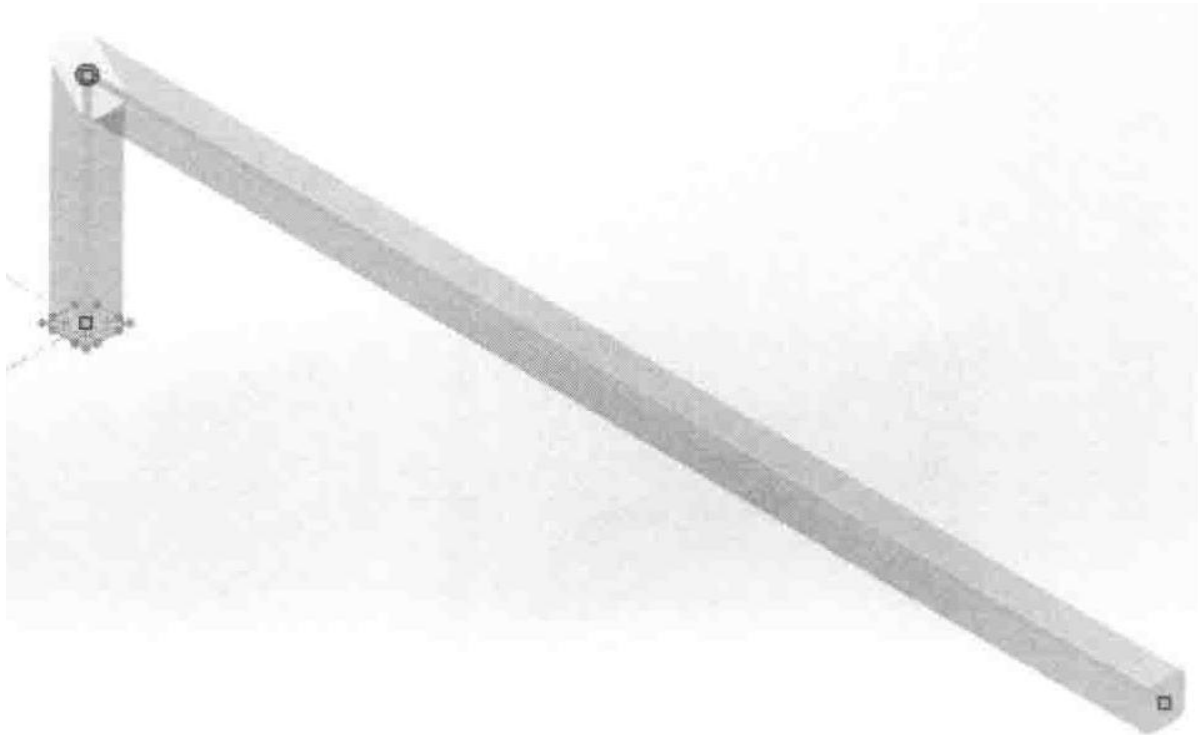
Problem 4.1

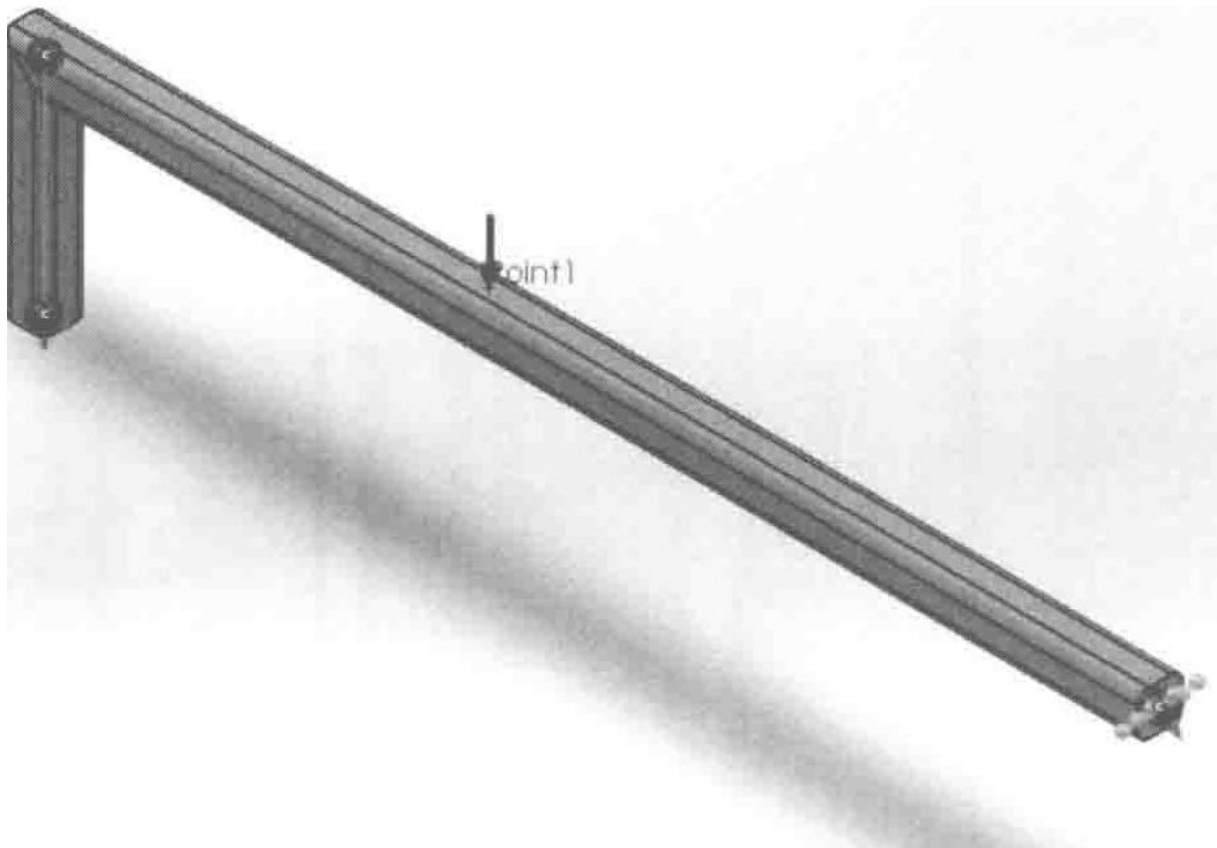
Solve for the forces at A and B of the beam shown in Figure



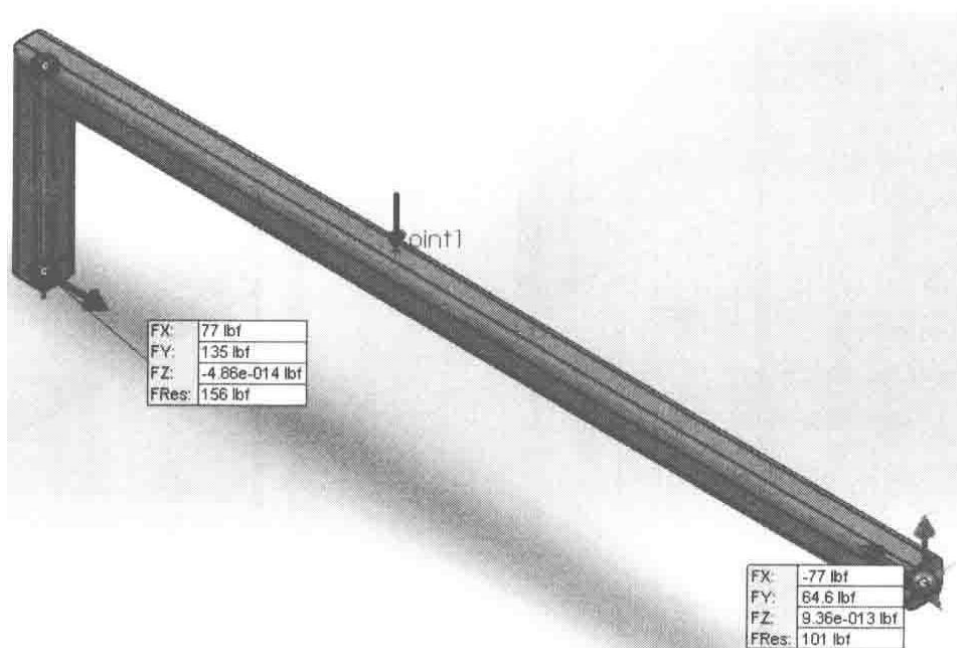
Solution:

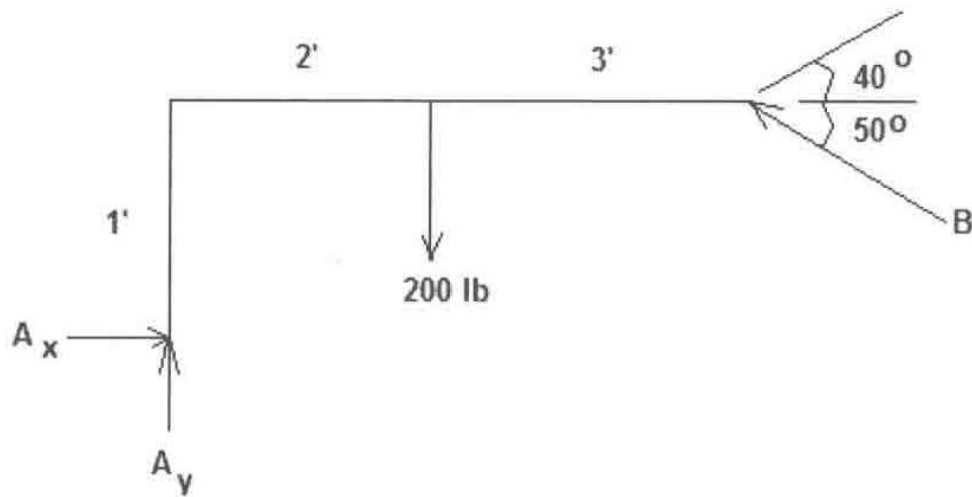






Result





FBD of the member

Validation

Draw the FBD as shown in Figure 4.15.

$$\sum M_A = 0; (B \cos 50)(1) + (B \sin 50)(5) - (200)(2) = 0$$

$$0.643B + 3.83B = 400$$

$$B = 89.4 \text{ lb } \underline{50^\circ} \uparrow$$

$$\sum F_x = 0; A_x - B \cos 50 = 0$$

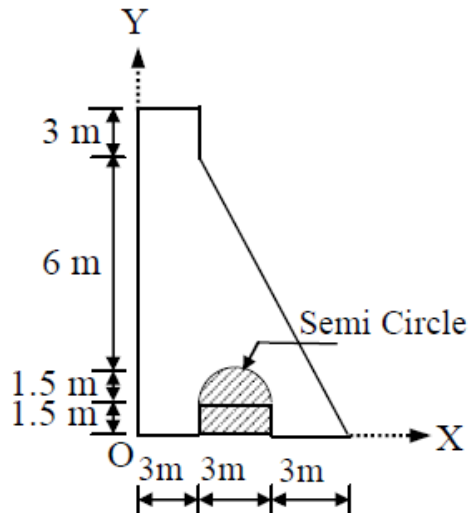
$$A_x = 57.5 \text{ lb } \rightarrow$$

$$\sum F_y = 0; A_y - 200 + B \sin 50 = 0$$

$$A_y = 132 \text{ lb } \uparrow$$

Centroid and Centre of Gravity

Problem: Determine the centroid of the composite area with respect to O shown in Fig below



Answer:

Let divide the given section in to 4 (four) part

- (1) : Rectangular (3 X 12)
- (2) : Triangle (6 x 9)
- (3) : Rectangular (3 x 1.5)
- (4) : Semi – circular (r = 1.5m)

Sr. no	Shape	Area (m ²)	x (m)	Y(m)	Ax (m ³)	Ay (m ³)
1	Rectangle	$A_1 = 12 \times 3 = 36$	$x_1 = \frac{3}{2} = 1.5$	$y_1 = \frac{12}{2} = 6$	$A_1 x_1 = 54$	$A_1 y_1 = 216$
2	Triangle	$A_2 = \frac{1}{2} \times 6 \times 9 = 27$	$x_2 = 3 + \frac{6}{3} = 5$	$y_2 = \frac{9}{3} = 3$	$A_2 x_2 = 135$	$A_2 y_2 = 81$
3	Rectangle	$A_3 = -3 \times 1.5 = -4.5$	$x_3 = 3 + 1.5 = 4.5$	$y_3 = \frac{1.5}{2} = 0.75$	$A_3 x_3 = -20.25$	$A_3 y_3 = -3.375$
4	Semi-circle	$A_4 = -\frac{\pi r^2}{2} = -3.53$	$x_4 = 3 + 1.5 = 4.5$	$y_4 = 1.5 + \frac{4r}{3\pi} = 2.134$	$A_4 x_4 = -15.885$	$A_4 y_4 = -7.53$

$$\bar{x} = \frac{\sum Ax}{\sum A} = \frac{A_1 x_1 + A_2 x_2 + \dots + A_n x_n}{A_1 + A_2 + A_3 + \dots + A_n} = 2.78 \text{ mm}$$

$$\bar{y} = \frac{\sum Ay}{\sum A} = \frac{A_1 y_1 + A_2 y_2 + \dots + A_n y_n}{A_1 + A_2 + A_3 + \dots + A_n} = 5.20 \text{ mm}$$

Software Solution:

1. Create the above sketch
2. Check for the section properties

Rectilinear and Angular motion

Introduction

Kinematics is the study of the geometry of motion without considering the forces that are causing the motion. It is concerned with quantities such as displacement, distance, velocity, and acceleration. There are several types of planar motions:

1. *Rectilinear or translational motion.* A particle moves in a straight line, without rotation.
2. *Circular motion.* A particle moves in a circular motion due to rotation.
3. *General plane motion.* A particle that may have both rectilinear and rotational motion simultaneously, as in the case of the link of slider crank mechanism.

What is rectilinear or translational kinematics?

Rectilinear (or translational) kinematics is the linear movement of a body without considering the forces that is causing the motion. Linear motion of a body is also called the rectilinear motion and is analyzed in terms of displacement, distance, velocity, and acceleration.

Linear Displacement

Displacement and distance are two different quantities. While distance is a scalar quantity, displacement is a vector quantity which has both magnitude and direction. Linear displacement is a vector quantity while linear distance is scalar quantity. The symbol used for linear displacement is s . s is measured either in meters, feet, or kilometers.

Linear Velocity

Linear velocity is the rate of change of linear displacement. It is represented by v . It is defined as $v = \frac{s}{t} = \frac{\Delta s}{\Delta t}$, where v is the average angular velocity, s is the linear displacement, and t is the time. The unit of v is m/s, km/h, ft/s, or ft/min and the direction must be given to complete the definition of velocity.

Linear Acceleration

Linear acceleration is the rate of change of linear velocity. It is represented by a and is defined as $a = \frac{v}{t} = \frac{\Delta v}{\Delta t}$, where a is the average linear acceleration, v is the linear velocity, and t is the time. The unit of a is m/s^2 , mm/s^2 , in/s^2 , or ft/s^2 .

Demonstration of Rectilinear and Angular Motion Using Pick & Place Robot

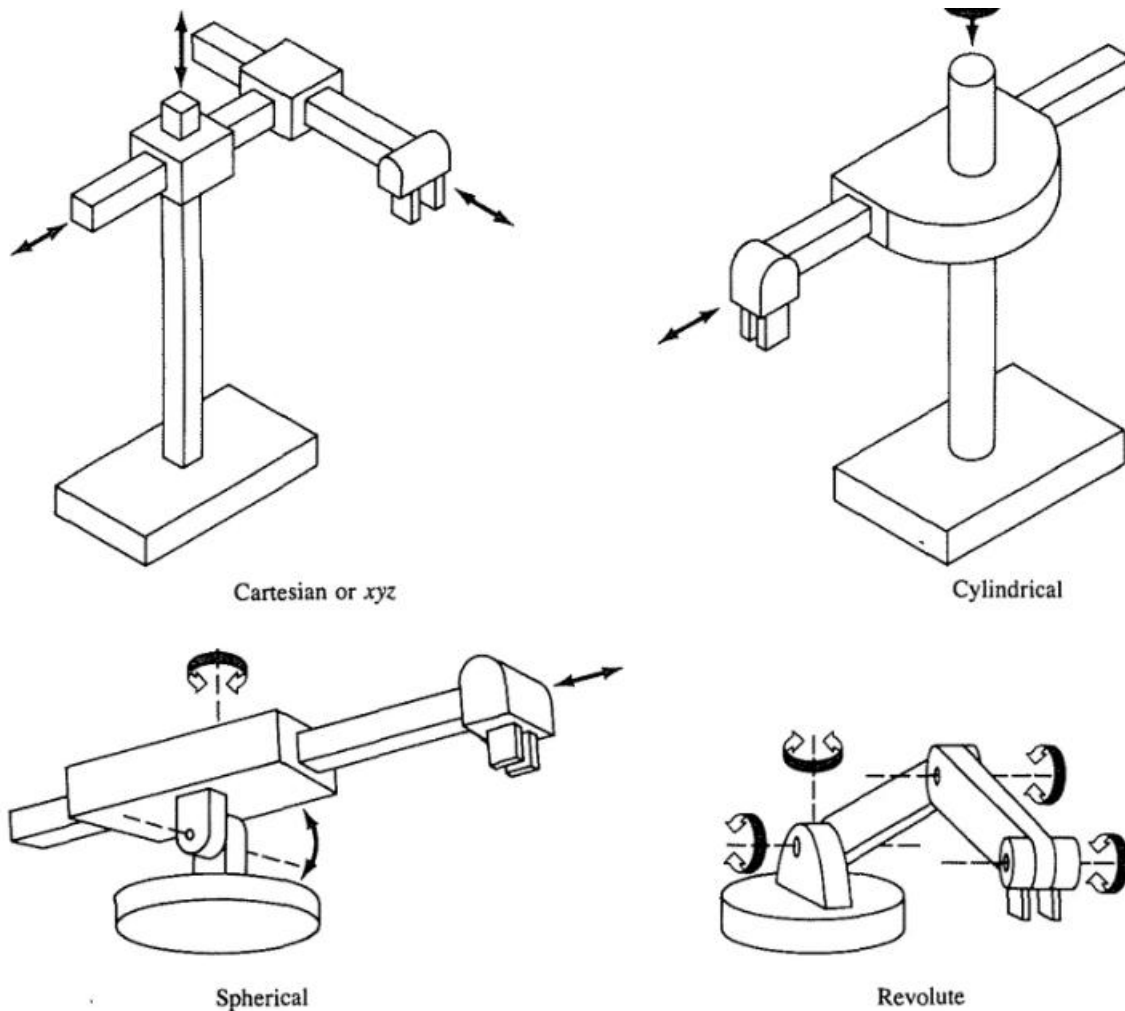


Fig: The four basic robot anatomies

Demonstration Procedure

Experiment 1: Horizontal Rectilinear Motion

Step1:

Program the robot to move only along the X-axis from Point A to Point B.

Step2:

Measure:

- Initial position (x_0)
- Final position (x)
- Time taken (t)

Step3:

Calculate:

Velocity $\rightarrow v = \frac{x-x_0}{t}$

If speed changes $\rightarrow$ Acceleration $\rightarrow a = \frac{v-u}{t}$

Experiment 2: Base Rotation Motion

Step1:

Rotate base from 0° to 90° .

Step2:

Measure:

- Angular displacement (θ)
- Time taken (t)

Step3:

Calculate:

- Angular velocity $\rightarrow \omega = \frac{\theta}{t}$
- Angular acceleration $\rightarrow \alpha = \frac{\omega-\omega_0}{t}$